# CHAPTER I

# INTRODUCTION

## Problem Background

The Sangguniang Kabataan (SK) plays a big role in supporting youth development and helping manage community activities at the barangay level. SK officials are responsible for planning programs, organizing events, and making sure that the needs of the youth are properly addressed. To carry out these responsibilities, they handle many important documents such as resolutions, activity proposals, accomplishment reports, meeting minutes, budgets, and approval forms. These documents serve as proof of transactions, references for decision-making, and records for accountability. Because of this, proper document management is very important for the smooth flow of SK operations.

However, in several barangays in Ormoc City specifically South, East, Ipil, Cagbuhangin, and Don Felipe Larrazabal most of these documents are still handled manually. Based on initial observations and informal conversations with SK officials, especially the SK Secretaries, documents are usually stored in folders, envelopes, cabinets, or sometimes even in boxes. While this method has been used for many years, it has become difficult to manage as the number of documents continues to grow.

In barangays like South and Cagbuhangin, where there are more activities and documents being produced, the lack of proper storage space makes it harder to keep records organized. Files are often stacked together, making it difficult to separate old and new documents. Over time, some documents get damaged due to moisture, dust, pests, or simple wear and tear. There are also instances where documents are misplaced or cannot be found when needed. Because of this, SK officials spend a lot of time just searching for specific files, which slows down their work and affects productivity.

Another issue is the absence of a centralized system. In barangays such as East and Ipil, documents are sometimes kept in different locations or handled by different individuals. This makes it harder to monitor both ongoing and completed activities. Documents that require multiple approvals often experience delays because they need to be physically passed from one person to another. SK Secretaries shared that this process can be stressful, especially when deadlines need to be met. In addition, there is no reliable tracking system to check the status of documents, which can lead to confusion or duplication of work.

Security is also a concern. Since the documents are stored physically, they can be accessed by unauthorized individuals or accidentally lost without proper monitoring. Important and sensitive information may be exposed, which can affect the credibility and reliability of SK records. Without a proper system in place, it becomes difficult to ensure that documents are safe and well-maintained.

Aside from document-related concerns, managing SK member information is also challenging. In barangays like Don Felipe Larrazabal and Ipil, there is no centralized record system for tracking member status. Information about whether a member is active, inactive, or no longer participating is not always updated. This creates confusion when preparing reports, monitoring attendance, or verifying participation in activities. It also affects decision-making, since accurate data is not always readily available.

Overall, these challenges greatly affect the efficiency and effectiveness of SK operations. Instead of focusing more on developing programs and serving the youth, SK officials spend a significant amount of time dealing with paperwork and organizing records. This slows down processes, reduces productivity, and can even affect the quality of services provided to the community. With the continuous advancement of technology, relying solely on manual systems is no longer practical or efficient.

Because of this, the study proposes the development of a digital document management and monitoring system in coordination with the SK Chairman of Ormoc City. The proposed system will allow authorized users to securely store, organize, and access documents in a centralized platform. It will also provide features for tracking document status, monitoring SK activities, and managing member information more efficiently. Users will be able to view SK accounts, access member lists, and check their current status (active, inactive, or quit) in real time.

In addition, the system is expected to reduce the time spent on searching and retrieving documents, minimize errors, and improve overall organization. It will also enhance data security by limiting access to authorized users only. By digitizing records and processes, the system will make SK operations faster, more accurate, and more reliable.

In the long run, the proposed system aims to help SK officials focus more on their primary goal, which is to serve the youth and the community. It will support better decision-making, improve transparency and accountability, and contribute to a more organized and efficient SK governance.

## Theoretical Background

The need for a more organized and systematic way of handling records becomes clear when looking at the current situation of the Sangguniang Kabataan (SK). As discussed in the introduction, the continued use of manual and paper-based processes makes it difficult to organize, retrieve, and monitor important documents. Because of these challenges, this study is anchored on Records Management Theory, which focuses on properly managing records throughout their lifecycle rom creation to storage and eventual disposal. Applying this theory helps guide the development of the proposed system, SKActHub, which aims to improve how SK records in Ormoc City are organized, accessed, and monitored.

### *Records Management Theory*

Records Management Theory explains that records should be handled in a structured and organized manner from the moment they are created up to the time they are no longer needed. This includes processes such as classification, storage, retrieval, and preservation. The theory views records as valuable assets because they serve as evidence of an organization's activities and decisions. According to the International Organization for Standardization (ISO 15489), effective records management ensures that records remain authentic, reliable, have integrity, and are usable over time. In simple terms, records should be trustworthy, protected from unauthorized changes, and easy to find when needed.

When records are not properly managed, several problems can arise. These include lost or duplicated documents, delays in decision-making, and difficulty in tracking important information. As explained by Shepherd and Yeo (2003), poor records management weakens institutional memory and reduces efficiency. In manual systems, records are also more likely to be damaged, misplaced, or accessed by unauthorized individuals. As the number of documents increases, these problems become more serious and can slow down operations.

In the context of the SK, records such as resolutions, activity proposals, project plans, accomplishment reports, approvals, financial documents, and correspondence are essential. These documents serve as proof of activities and are needed for reporting, auditing, and transparency. However, because these are mostly stored physically, SK officials often experience difficulty in retrieving documents, monitoring activities, and verifying past decisions. This clearly shows the need for a more efficient system.

This is where Records Management Theory strongly supports the development of SKActHub. The proposed system follows the principles of the theory by digitizing records and organizing them in a structured way. Through features such as categorization, tagging, and indexing, SKActHub makes documents easier to find and manage. Instead of manually searching through folders, users can quickly retrieve files within seconds.

The theory also emphasizes the importance of managing the lifecycle of records. SKActHub supports this by allowing documents to move step-by-step from creation and submission to review, approval, storage, and monitoring. This means that documents are not just stored but are actively managed throughout their use. This directly solves the problem of scattered and untracked documents in the current system.

Another important principle of the theory is security and control. SKActHub applies this by implementing role-based access, where the SK Chairman can view all documents, while the Secretary and Treasurer can only access the files they uploaded or the folders they created. This ensures data confidentiality while still allowing collaboration within the system. Backup features also help protect records from loss or damage.

Records Management Theory also supports transparency and accountability. With organized and accessible records, SK officials can easily prepare reports, respond to inquiries, and provide documentation when needed. This improves trust and ensures that all activities are properly recorded and traceable.

In addition, digital records management helps maintain continuity of operations. According to Penn, Pennix, and Coulson (1994), well-managed records ensure that organizations can continue functioning effectively even during changes in leadership. In the case of the SK, this means that new officers can easily access past documents, project histories, and reports. This prevents the loss of important information and supports smoother transitions.

Overall, Records Management Theory strongly supports this study because it provides the foundation for organizing, securing, and managing records efficiently. By applying this theory through SKActHub, the system not only addresses the current problems in document handling but also improves overall efficiency, transparency, and decision-making within the SK.

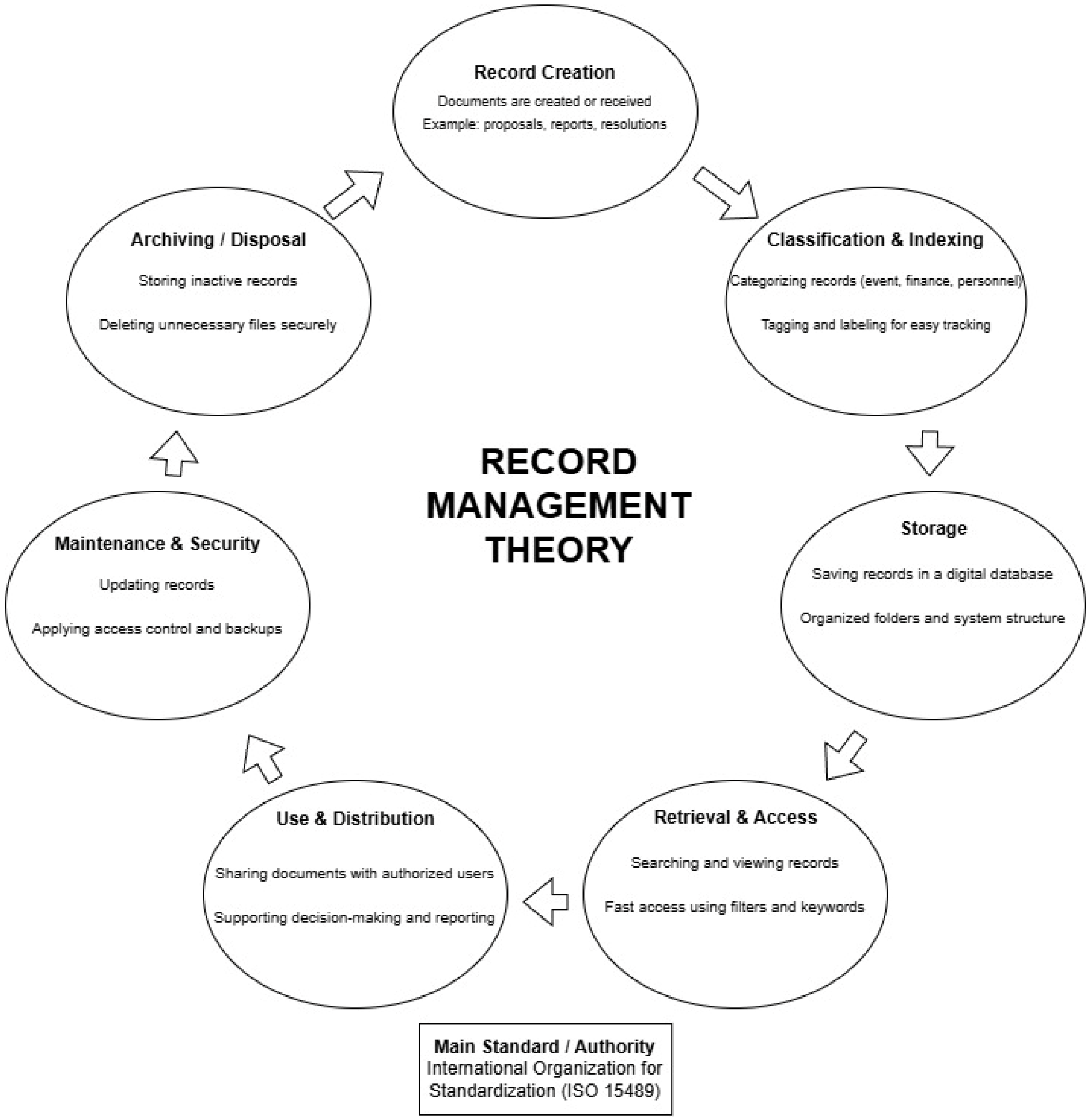


***Figure 1: Record Management Theory***

***Figure 1*** Illustrates the application of Records Management Theory in the proposed system through a continuous lifecycle process. It shows how records are created, classified, stored, retrieved, used, maintained, and eventually archived or disposed of in a structured and organized manner. Each stage ensures that records remain accurate, secure, and easily accessible throughout their lifecycle. The system supports efficient document handling by enabling quick retrieval, controlled access, and proper storage, while also maintaining data integrity through security and backup mechanisms. This process allows the organization to effectively manage information, support decision-making, and ensure accountability and transparency in all operations.

### *Conceptual Framework*

This framework demonstrates how the system transforms manual, paper-based operations into a centralized, automated, and secure digital environment. It emphasizes the flow of information from user-provided data and system interactions to processed outputs ensuring transparency, accountability, and operational efficiency within the Sangguniang Kabataan offices of selected barangays in Ormoc City.

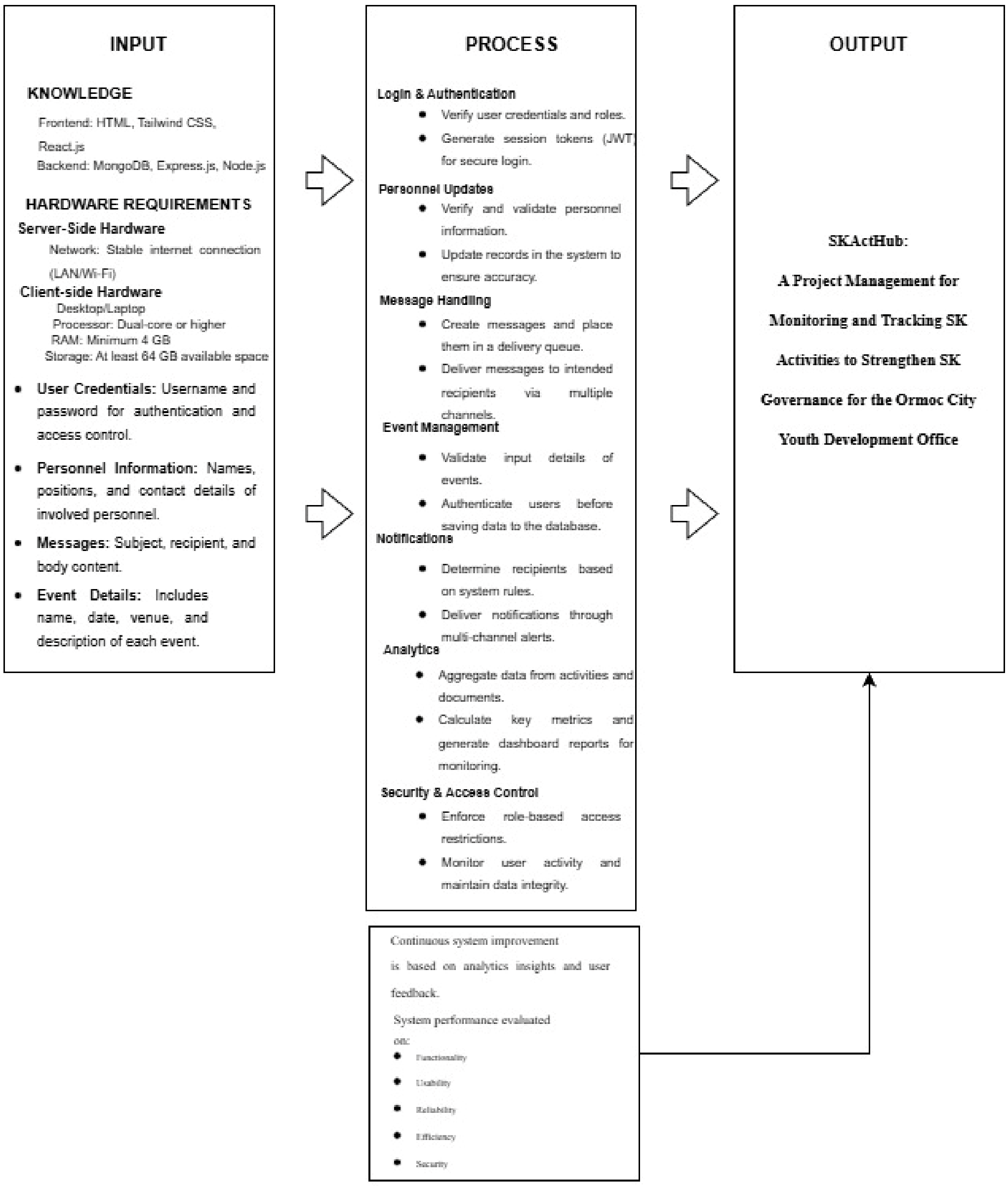

*Figure 2 : Conceptual Framework*

***Figure 2*** presents the Input–Process–Output (IPO) model of the proposed SKActHub system. The input stage includes knowledge resources, hardware requirements, user credentials, personnel information, messages, and event details necessary for system operation. These inputs are processed through key system functions such as login and authentication, personnel updates, message handling, event management, notifications, analytics, and security with access control. Each process ensures that data is validated, organized, and protected. As a result, the system produces outputs in the form of a fully functional SKActHub platform that supports the monitoring and tracking of SK activities, promotes efficient governance, and enhances transparency and decision-making within the Ormoc City Youth Development Office.

**Statement of the Problem**

The Sangguniang Kabataan (SK) plays an important role in youth governance and in implementing programs within the community. To perform these responsibilities effectively, proper management of documents and member records is necessary to ensure transparency, accountability, and smooth operations. However, many SK offices in Ormoc City still rely on manual and paper-based systems. Because of this, documents are often lost, misplaced, or duplicated, and retrieving important files can take a lot of time. It is also difficult to monitor the status of submitted proposals and activities, especially when multiple steps and approvals are involved.

Given these challenges, this study seeks to answer the following questions:

- What kind of web-based system can be developed to better organize, store, archive, and manage SK documents and member information?

- What features should be included to make uploading, categorizing, searching, and retrieving documents easier and faster?
- How can the system help track the status of documents from submission to review and approval to improve workflow monitoring?
- How can a centralized account management feature help authorized users monitor the status of SK members?.
- What security measures, such as user authentication and role-based access control, are needed to protect sensitive documents and records?
- How effective is the proposed system in terms of improving the accessibility of data, organization and storage of documents, monitoring and tracking of activities, security and protection of records, and overall usability for SK officials?

**Objectives of the Study**

**Project Goal**

The main goal of this study is to design and develop SKActHub, a web-based Digital Document Management, Archiving, and SK Account Management System. The system aims to improve the efficiency, security, accessibility, and monitoring of official documents and SK member records for the Sangguniang Kabataan of Ormoc City. To achieve this goal, the study pursues the following objectives:

- To design and develop a web-based system that can organize, store, archive, and manage SK documents and member information efficiently

- To develop system features that enable users to upload, categorize, search, and retrieve documents efficiently, while ensuring that all documents undergo an approval process before final storage.
- To implement a document tracking system that monitors the status of documents from submission to review and approval.
- To develop a centralized SK account management module that enables authorized users to view all members and monitor their status.
- To incorporate security features such as user authentication, role-based access control, and data protection to ensure the safety of documents and records.
- To evaluate the effectiveness of the system in terms of usability, efficiency, accessibility, security, and accuracy in managing documents and member information.

**Significance of the Study**

**For the Sangguniang Kabataan (SK) Officials of Ormoc City**

The proposed system, SKActHub, is developed to enhance the efficiency, organization, and overall productivity of SK officials in Ormoc City. By transitioning from manual and paper-based processes to a centralized digital platform, the system enables more systematic storage, management, and retrieval of documents. This reduces the time and effort required to locate files and organize records, allowing officials to focus more on planning, implementing, and monitoring youth-related programs. Furthermore, the system supports improved workflow through real-time tracking of documents and activities, contributing to a more streamlined and effective governance process.

### For the SK Chairperson, Secretary, and Treasurer

SKActHub provides reliable and well-structured information that supports informed decision-making among key SK officials. The Chairperson, Secretary, and Treasurer can efficiently review proposals, monitor accomplishment reports, and oversee the status of document approvals. Additionally, the system offers a centralized database of SK member records, enabling easy tracking of member status, whether active, inactive, or no longer in office. This ensures greater accuracy in record-keeping, minimizes errors, and promotes transparency and accountability within the organization.

### For the Ormoc City Youth Development Office (OCYDO)

The Ormoc City Youth Development Office (OCYDO) will benefit from improved access to organized and digitized SK records through the implementation of SKActHub. The system enables faster and more convenient review of proposals, reports, and other documents submitted by various SK offices. As a result, the YDO can more effectively monitor youth programs, ensure compliance with established guidelines, and make data-driven decisions based on accurate and up-to-date information.

### For Future Researchers

This study serves as a valuable reference for future researchers who aim to develop digital solutions for local government units and similar institutions. It offers insights into the design, development, and implementation of a web-based document management and monitoring system. Moreover, it provides a foundation for further innovation, as future researchers may enhance the system by integrating advanced features, improving functionality, or expanding its application to other areas of digital governance and records management.

### Scope and Delimitations of the Study

This study focuses on the design and development of SKActHub, a web-based Project Management, Document Management, and SK Account Management System intended to support the monitoring and tracking of Sangguniang Kabataan (SK) activities and SK membership information in order to strengthen SK governance for the Ormoc City Youth Development Office.

### Scope

This study focuses on the design, development, and implementation of SKActHub for the Sangguniang Kabataan (SK) of Ormoc City. The system is intended to support the efficient management, archiving, organization, retrieval, and monitoring of documents related to SK activities. These documents include activity proposals, accomplishment reports, approvals, budgets, minutes of meetings, financial reports, resolutions, and other official SK records.

The implementation and testing of the system are limited to selected barangays in Ormoc City, specifically Barangay South, Barangay East, Barangay Ipil, Barangay Cagbuhangin, and Barangay Don Felipe Larrazabal. These barangays serve as the primary participants and pilot users of the system, allowing for proper evaluation and validation of its functionality and effectiveness.

In addition, the system includes a centralized SK account and membership management feature. This allows authorized users to view SK accounts, access a complete list of SK members, and monitor each member's status whether active, inactive, or no longer part of the organization. This feature aims to improve accuracy in record-keeping and enhance administrative monitoring.

SKActHub implements a role-based access system intended for the SK Chairperson, SK Secretary, and SK Treasurer. Each user is provided with an individual account. The SK Chairperson has full access to all documents and system features, while the SK Secretary and SK Treasurer can manage and access only the documents they have personally submitted, based on their assigned responsibilities.

SKActHub mainly focuses on document posting, project and activity monitoring, account viewing, and membership tracking. It enables authorized users to upload, store, organize, retrieve, and monitor various documents related to SK activities while keeping member records updated. Through these features, the system aims to improve the monitoring of submitted documents, completed activities, and the participation of SK members.

Lastly, the system is developed as a web-based application using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js, based on the developers' technical knowledge and expertise. Since it is web-based, the system is designed to be accessed through a web browser and requires a desktop or laptop computer with a stable internet connection.

**Delimitations**

This study focuses on the development and implementation of SKActHub for the Sangguniang Kabataan (SK) of Ormoc City, specifically covering five selected barangays: Barangay South, Barangay East, Barangay Ipil, Barangay Cagbuhangin, and Barangay Don Felipe Larrazabal. It does not include all SK offices in Ormoc City, as doing so would be too broad for a capstone project. Instead, the selected barangays provide a manageable scope that still reflects the actual needs and processes of SK offices, allowing the researchers to effectively develop and evaluate the system.

SKActHub is developed as a web-based system intended for use on desktop and laptop computers through a web browser. It is not optimized for mobile devices, which may limit accessibility for users who rely on smartphones. The system implements role-based access for key SK officials, namely the SK Chairperson, SK Secretary, and SK Treasurer, all of whom are provided with individual accounts.

The SK Chairperson is granted full access to all documents within the system, allowing comprehensive oversight and management. In contrast, the SK Secretary and SK Treasurer have limited access, where they can only view, manage, and retrieve the documents they have personally submitted or stored. This role-based structure ensures proper control, maintains data security, and clearly defines responsibilities among users while supporting efficient system management.

The system mainly focuses on document management, activity monitoring, and tracking the status of SK members. It does not include more advanced features such as automated approval workflows, financial tracking systems, attendance monitoring, role-based access control beyond the defined users, performance evaluation tools, offline functionality, real-time analytics, or integration with other government information systems. These limitations are intentional, as the system is designed to prioritize essential features that support efficient record-keeping and basic operational monitoring rather than complex, enterprise-level automation.

**Definition of Terms**

- ***Centralized SK Management Platform* -** A digital platform utilized by SK officials in selected barangays to upload, organize, monitor, and manage activity proposals, accomplishment reports, and member information efficiently.
- ***Participating SK Offices* -** The SK offices in selected barangays of Ormoc City that use SKActHub to manage documents and monitor youth membership records.
- ***System Administrator* -** The primary user of SKActHub in each barangay who uploads documents, monitors activities, manages membership status, and controls access permissions.
- ***Document and Records Assistant* -** A secondary user of SKActHub who supports the SK Chairman by uploading documents, updating member information, and maintaining activity records.
- ***Financial Records Handler* -** A secondary user of SKActHub who supports the SK Chairman by managing financial records, uploading financial documents, and maintaining budget and expense reports related to SK activities.
- ***Participation Classification Indicator* -** The designation used by SK officials to track member participation and eligibility in programs and activities.
- ***Planned Activity Record* -** A proposal prepared and submitted by the SK to document planned youth programs or events for monitoring and reference.
- ***Completed Activity Record* -** A report used by SK officials to document completed programs and evaluate activity performance.
- ***Document Handling Feature* -** The feature within SKActHub that enables SK officials to manage and access digital documents efficiently.

- ***Document Handling Feature*** **-** A system feature that allows the SK Chairman to assign and manage access privileges of the SK Secretary and SK Treasurer within SKActHub. The Chairman can monitor and access all uploaded documents, whereas the Secretary and Treasurer are limited to viewing and managing only the documents they submitted or the folders they created.
- ***Browser-Based System Interface*** **-** The interface of SKActHub accessed via desktop or laptop devices where users perform document management and monitoring tasks.
- ***MERN Stack*** **-** The technological foundation of SKActHub that enables data storage, system processing, and interactive dashboard functionality.
- ***Submission and Tracking Process*** **-** The process used by SK officials to ensure that proposals and reports are properly recorded, updated, and accessible.

# CHAPTER II

## Review of Related Literature

Effective document management and activity monitoring play a vital role in promoting transparency, accountability, and operational efficiency in institutions such as the Sangguniang Kabataan (SK). Proper record-keeping ensures that programs, financial transactions, and official activities are well-documented, providing reliable references that support informed decision-making and strengthen public trust. However, many SK offices still rely on traditional paper-based filing systems, which often result in misplaced documents, delayed retrieval, duplication, and limited accessibility. These manual processes also increase the risk of errors and physical damage to records, making it more difficult to monitor activities efficiently.

In many cases, SK officials spend unnecessary time searching for documents or verifying records, which slows down daily operations and affects productivity. This situation can also lead to inconsistencies in reporting and challenges in tracking the progress of activities. As a result, the overall management of records and monitoring of programs becomes less organized and more prone to inefficiencies.

Studies show that digital document management systems significantly improve record organization, security, and accessibility. Centralized platforms allow faster document retrieval, better classification, and more secure storage of files. Features such as automated tracking and role-based access control further enhance transparency by ensuring that only authorized personnel can access sensitive information. Despite these advantages, there are still limited studies focused on integrated systems specifically designed for SK operations, highlighting the need for a platform that combines document management and activity monitoring to improve administrative efficiency.

## Local Studies

### Digital Document Management Systems

Dizon and Cruz (2023) developed a blockchain-based barangay document management system integrated with Optical Character Recognition (OCR) technology in a Philippine local government office. The study included 50 barangay staff members who utilized the system for six months. Their findings revealed that automation reduced manual encoding errors by 40%, accelerated document processing by 30%, and improved record immutability through blockchain technology, preventing unauthorized alterations. The study emphasizes that the integration of emerging technologies enhances public trust, transparency, and data integrity within local government operations, supporting the rationale for SKActHub's secure storage design.

Similarly, Innovatus Research Group (2019) implemented a web-based document management system in a barangay office in Leyte. Using pre- and post-implementation assessments, they found notable improvements in document classification, retrieval speed, and inter-office coordination. Document retrieval time was reduced from an average of 30 minutes under a paper-based system to under 5 minutes with the digital system. The study highlights that digital repositories reduce physical storage dependency and improve administrative efficiency, directly supporting SKActHub's document storage and organization functionality.

Reyes et al. (2021) studied cloud-based document archiving in municipal offices and reported enhanced compliance with record-keeping policies, reduced document misplacement by 65%, and faster access to records. The study reinforces the importance of centralized document repositories, which SKActHub provides for SK activity proposals, accomplishment reports, and membership data.

**Document Tracking and Workflow Monitoring**

Bautista et al. (2022) developed a web-based document tracking system for a Philippine institutional setting involving 75 employees over six months. The system allowed real-time monitoring of documents from submission to approval, incorporating automated status updates, email and SMS notifications, and visual tracking dashboards. The study reported a 35% reduction in processing delays, improved coordination among departments, and faster completion of multi-step approval workflows. These findings support the inclusion of SKActHub's automated status updates and notification modules to improve the efficiency of SK document handling.

Similarly, Lopez and Mendoza (2023) implemented a barangay request and tracking system in Leyte, designed to minimize human errors, generate automated performance indicators, and enhance transparency. Their evaluation indicated that the system increased accountability as each document could be traced from submission to completion, reducing misplaced files and confusion during approval.

Santos et al. (2021) examined workflow monitoring in several Philippine municipal offices. Their research indicated that real-time dashboards and automated alerts facilitated faster decision-making, improved compliance with reporting timelines, and allowed administrators to detect bottlenecks in activities. These studies collectively validate SKActHub's inclusion of document tracking and workflow monitoring as essential components for operational transparency and administrative efficiency.

## Cloud Computing in Document Management

Garcia and Lim (2020) explored the adoption of cloud-based document management systems in government offices and found that cloud technology significantly reduced infrastructure and maintenance costs. Their study showed that organizations no longer needed extensive physical storage or on-site servers, as documents could be securely stored in remote servers. This allowed employees to access files anytime and anywhere, improving flexibility and productivity. The system also enhanced collaboration, as multiple users could work on the same document without the need for physical file transfers.

## Information Systems in Local Governance

Navarro and Perez (2022) examined the implementation of integrated information systems in Philippine local government units and found that these systems significantly improved coordination between departments and reduced delays in document processing. Their study demonstrated that centralized platforms enable data consolidation, allowing multiple offices to access the same information in real time. This reduced redundancy in records and minimized inconsistencies in reporting. Furthermore, the system improved workflow efficiency by eliminating repetitive manual encoding and simplifying data retrieval processes, which contributed to faster administrative operations.

Villanueva et al. (2023) further supported these findings by analyzing the use of information systems in municipal offices. Their study revealed that systems with centralized dashboards improved monitoring capabilities by allowing administrators to track activities, documents, and performance indicators in a single interface. This reduced the need for manual follow-ups and improved communication between departments. The researchers concluded that integrated systems enhance operational efficiency and accountability, particularly in environments where multiple processes occur simultaneously.

## Local Literature

Torres (2021) emphasized that information systems contribute to evidence-based governance by providing accurate, timely, and accessible data. The study highlighted that real-time information allows government officials to monitor ongoing activities, evaluate program effectiveness, and make informed decisions. In the context of youth governance, such as the Sangguniang Kabataan, the availability of reliable data ensures that projects and activities are properly documented and evaluated. The use of digital systems also improves transparency, as records can be easily reviewed and verified when needed.

## Foreign Studies

### Digital Document Management Systems

and Fianty (2023) investigated web-based document management systems in multiple Southeast Asian municipalities, involving 100 administrative users. Their study highlighted that centralized storage, workflow automation, and role-based access control significantly improved operational efficiency and interdepartmental collaboration. Audit trail features enabled supervisors to monitor document modifications, access history, and user activities, which increased transparency and minimized disputes between departments. The researchers also emphasized that digital systems reduced redundancies caused by duplicate records and improved reporting accuracy by providing real-time updates on the status of requests and documents. These findings demonstrate that integrating workflow automation and role-based permissions can enhance accountability, a feature directly relevant to SKActHub's design.

Similarly, Smith et al. (2022) conducted a European study on local government digital archives, analyzing the effects of digital systems on record retrieval and data security. Their study found that digital archiving reduced document retrieval time by 60% and enhanced compliance with international data security standards, such as GDPR. The system also included version control, which prevented accidental or unauthorized modifications, ensuring document integrity. These findings highlight the importance of secure, centralized storage and automated access control in managing sensitive records.

**Document Tracking and Workflow Monitoring**

Awang Gani et al. (2024) investigated electronic document management systems in Malaysian government agencies. Their study highlighted that workflow monitoring strengthened compliance with institutional procedures, reduced the risk of lost documents, and improved audit readiness. The system's automated logging mechanisms provided verifiable trails for all activities, ensuring accountability and simplifying audits.

O'Connor and Lee (2022) studied digital workflow systems in Canadian municipal offices, observing that integrated dashboards and alerts facilitated task tracking, improved coordination across departments, and enabled administrators to detect and resolve process delays early.

## Cloud Computing in Document Management

Chen et al. (2021) investigated cloud-based systems in government institutions and highlighted their role in disaster recovery and data protection. Their findings revealed that automatic backup features ensured that documents were preserved even in cases of hardware failure, natural disasters, or accidental deletion. This capability is particularly important for government offices that rely heavily on documentation, as it prevents permanent data loss and ensures continuity of operations.

Alotaibi and Federico (2021) examined the security aspects of cloud-based document management systems and found that features such as encryption, multi-factor authentication, and role-based access control significantly improved data protection. Their study reported a decrease in unauthorized access incidents and increased user confidence in the system. These security measures ensure that only authorized personnel can access sensitive information, which is critical in managing government records.

Rahman et al. (2023) also highlighted that cloud-based systems improve system scalability, allowing organizations to expand storage capacity without major infrastructure changes. Their study showed that scalable systems can accommodate increasing volumes of data, making them suitable for growing institutions. This flexibility ensures that document management systems remain efficient even as organizational needs evolve.

# CHAPTER III

## Methodology

Developing an effective technological solution requires a clear and well-organized approach. This chapter presents the methodology used in this study, including the processes involved in designing, developing, and evaluating SKActHub. It explains the research framework, system development approach, and the steps followed to ensure that the system addresses the challenges faced by the Sangguniang Kabataan in managing documents, monitoring activities, and tracking membership records. By following a structured method, the study aims to produce a system that is practical, reliable, and aligned with the needs of SK officials in the selected barangays of Ormoc City.

### Project Design

This study used the Developmental Research Method, which focuses on the systematic design, development, implementation, and evaluation of a system to address specific organizational needs. Unlike descriptive research, which mainly describes existing conditions, developmental research aims to create and validate a technological solution that can improve current processes. This approach is suitable for information technology projects because it provides a clear and structured way of turning identified problems into a functional digital system.

In this study, the method guided the development of SKActHub by first identifying the challenges in manual document handling, activity monitoring, and membership record management within the Sangguniang Kabataan. These identified issues served as the foundation for defining the system's objectives, features, and overall design.

The main goal of the study is to develop SKActHub, a web-based system for project management, document management, and SK account management for selected barangays in Ormoc City. The development process followed key stages such as planning, requirements analysis, system design, development, testing, and deployment to ensure that the system meets both user needs and operational requirements. The system is designed to provide secure document storage, organized record management, activity tracking, and centralized monitoring of SK members. Before implementation, the system was carefully tested and improved to ensure its functionality, usability, reliability, and security.

**Development Model**

The development of the system followed the Software Development Life Cycle (SDLC) to ensure a structured and organized process. In this study, the Agile Development Model was used because it allows flexibility, continuous feedback, and gradual improvement while the system is being developed.

The process started with gathering and analyzing requirements. During this stage, the researchers identified the needs of SK officials and reviewed how documents are currently managed in the selected barangays. This helped determine the important system features, types of documents to be handled, user roles, and the requirements for monitoring activities. Based on this information, the system design was created, including the overall structure of the system, database design, user interface, and workflow. Key features such as user accounts, document management, activity monitoring, and notifications were planned at this stage.

After the design phase, the system was developed using the MERN stack, which includes MongoDB, Express.js, React.js, and Node.js. The development was done step by step, starting with core features like user authentication, document uploading, and activity management. Additional features such as notifications and dashboards were added afterward. This gradual approach made it easier to improve and adjust the system as development progressed.

Once development was completed, testing was conducted to ensure that the system worked properly and was easy to use. Functional testing and usability testing were performed to identify errors and areas that needed improvement. Any issues found were fixed, and changes were made based on feedback from potential users.

After testing, the system was deployed and implemented in selected barangays for actual use and evaluation. Finally, the system underwent continuous monitoring and improvement. The researchers observed its performance, addressed any issues encountered, and made further enhancements based on user feedback. This ongoing process helped ensure that the system remained effective, reliable, and responsive to the needs of its users.

## Entity Relationship Diagram (ERD)

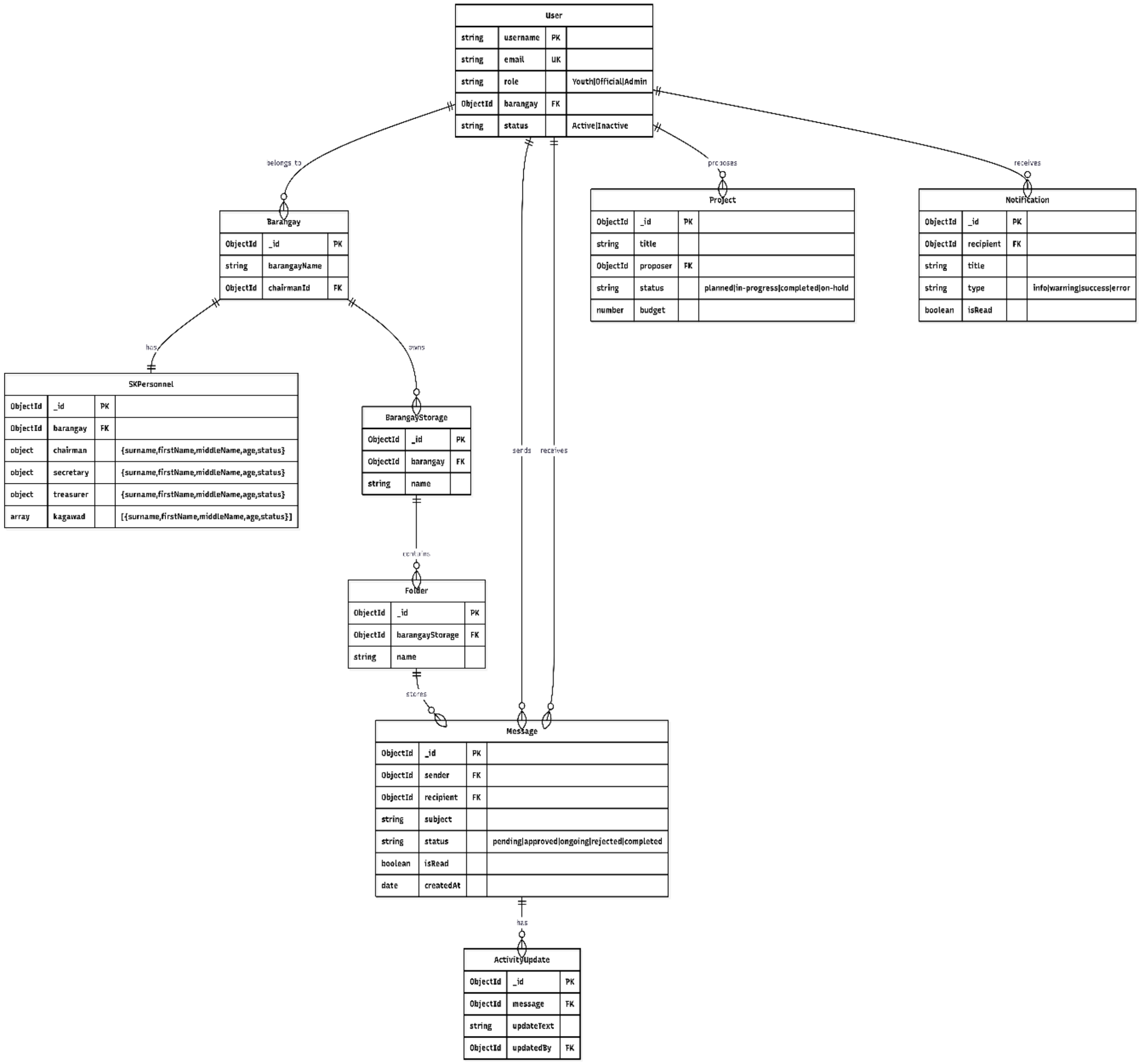


***Figure 3: Entity Relationship Diagram***

***Figure 3*** shows the overall structure of the SKActHub system database and how its main entities are connected. It highlights key components such as User, Barangay, SK Personnel, Projects, Messages, Notifications, and Storage, which work together to manage records, communication, and activities. The diagram illustrates how data flows within the system, ensuring organized information handling and efficient operations for SK officials.

## DATA FLOW DIAGRAM

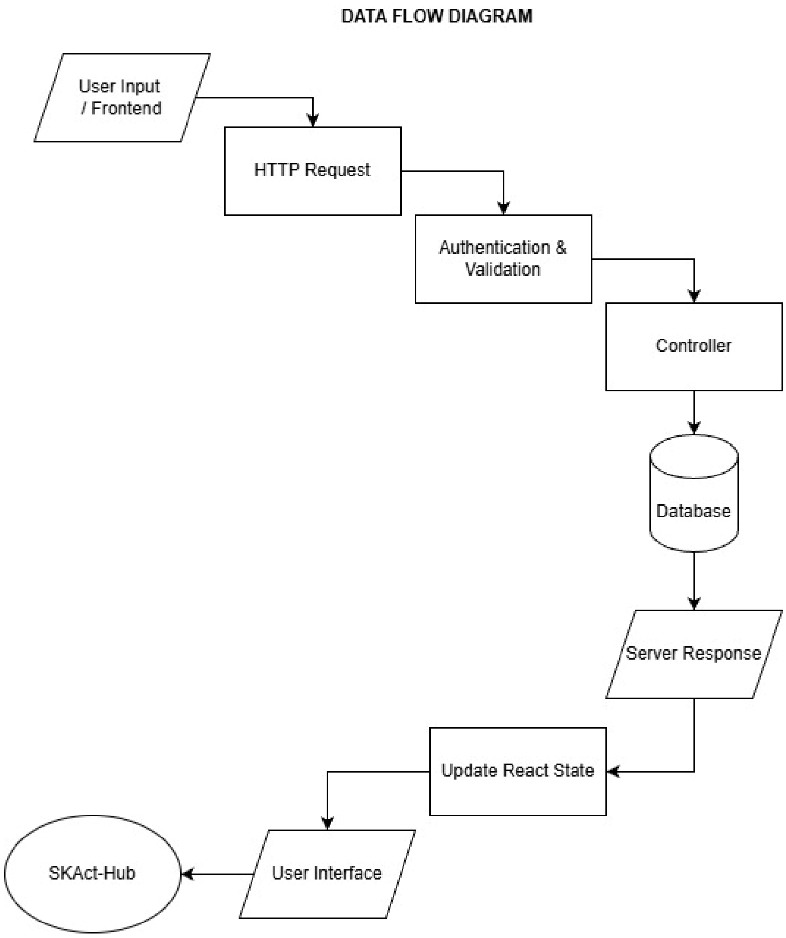


***Figure 4: Data flow Diagram for SKAct-Hub***

*Figure 4* This figure illustrates the flow of data within the SKAct-Hub system. It begins with user input from the frontend, which is sent as an HTTP request to the server. The request then undergoes authentication and validation to ensure security and correctness of the data. Once validated, the request is processed by the controller, which interacts with the database to store or retrieve necessary information. After processing, a server response is generated and sent back to the frontend, where the React state is updated accordingly. Finally, the updated data is reflected in the user interface, allowing users to view the results of their actions within the system.

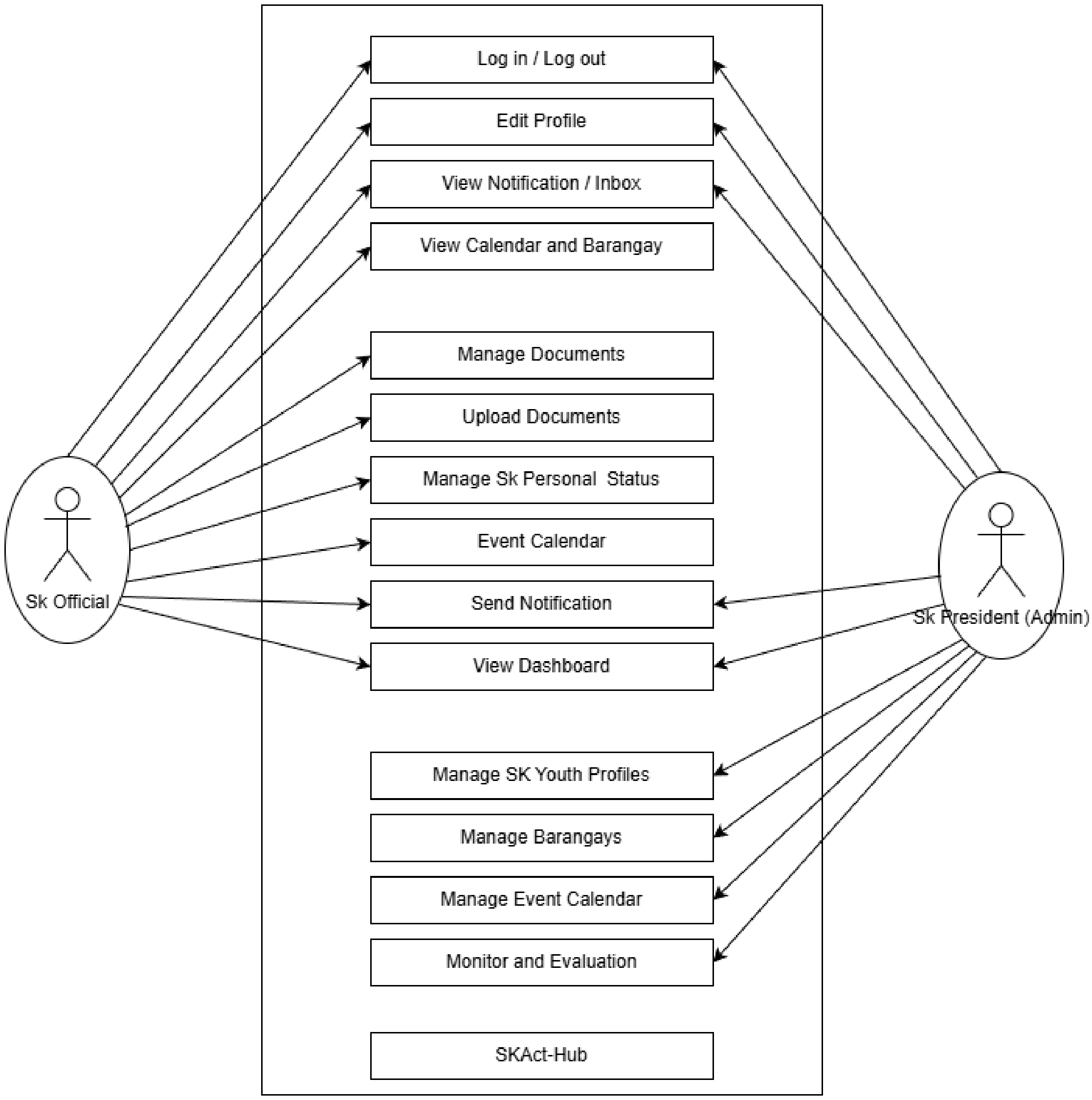


***Figure 5: Use Case Diagram for SKAct-Hub***

*Figure 5* illustrates the interactions between the SK Official and SK President (Admin) with the SKAct-Hub system. It shows the main functions available in the system such as login/logout, managing documents, viewing notifications, event calendar, and dashboard. The diagram also highlights the additional administrative functions of the SK President, including managing youth profiles, barangays, and monitoring activities.

***System Architecture***

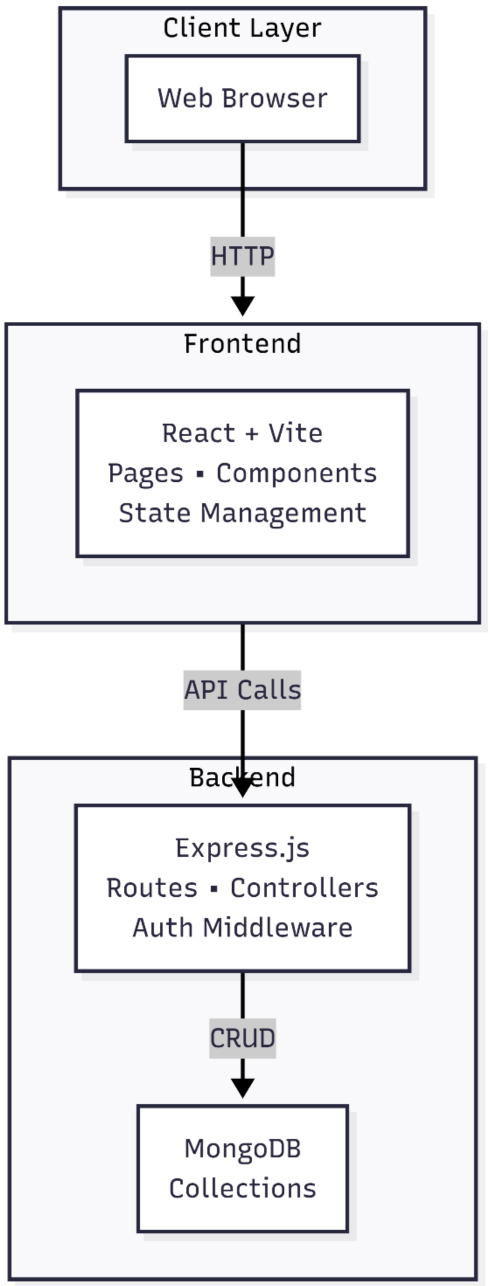


***Figure 6: System Architecture***

*Figure 6* shows how the different parts of the system work together in a simple and connected flow. The process starts when the user opens the system in a web browser, which then interacts with the frontend built using React and Vite to display pages and handle user actions. The frontend sends requests to the backend using API calls, where Express.js processes the data, applies security checks, and communicates with the MongoDB database to store and retrieve information.

### *A.Project Locale*

This study was conducted in five (5) selected barangays in Ormoc City, Leyte, Philippines. Ormoc City is a component city located in the western part of Leyte Province in Eastern Visayas, known for being an important center for local governance, business, and community activities. The barangays included in this study are South, East, Ipil, Cagbuhangin, and Don Felipe Larrazabal. These areas were chosen because they are accessible to the researchers, have active Sangguniang Kabataan (SK) officials, and still rely mostly on manual or paper-based systems in managing their documents.

In these barangays, SK officials usually keep records such as activity reports, financial documents, meeting minutes, and member information in folders, notebooks, or filing cabinets. While this method has been used for a long time, it often leads to misplaced documents, difficulty in finding records, and delays in completing tasks. It can also make it harder for officials to track activities and stay organized, especially when handling multiple responsibilities.

Because of these challenges, this study aims to introduce the SKActHub system, which is designed to make document management easier and more efficient. By using this system, SK officials can store, organize, and monitor their records digitally, making information easier to access and manage. In the long run, this can help improve coordination among officials, speed up administrative work, and support better service for the youth in these communities.

### *B.Respondents of the Project*

The respondents of the study are the official members of the Sangguniang Kabataan (SK) from five selected barangays. This includes five (5) SK Chairpersons, five (5) SK Secretaries, five (5) SK Treasurers with each barangay represented by its respective set of officials. These participants were chosen because of their direct involvement in SK operations and document management within their communities.

The SK Chairperson serves as the leader of the organization and is responsible for overseeing youth programs and activities. Their role includes presiding over meetings, approving proposals and reports, and ensuring that all projects are properly implemented in accordance with existing rules and regulations. Meanwhile, the SK Secretary is primarily responsible for handling documentation and records, such as preparing meeting minutes, organizing proposals and accomplishment reports, and maintaining files to ensure that all documents are properly recorded and easily accessible. The SK Treasurer, on the other hand, manages the financial aspects of the SK, including handling budgets, recording financial transactions, and preparing financial reports to ensure proper use of funds.

In addition, the SK Chairperson play an important role in supporting the implementation of SK programs and activities. They assist in planning and executing projects, participate in decision-making, and help monitor ongoing activities within the barangay. Their involvement also includes contributing to the preparation and submission of necessary documents and reports.

These respondents were selected because they are directly involved in the preparation, submission, organization, and monitoring of SK-related documents and activities. Their roles provide them with firsthand experience of the current processes used in their barangays. As a result, their feedback is essential in evaluating the effectiveness, usability, and overall functionality of the proposed SKActHub system.

### *C. Determination of the Sample Size*

The study involved a total of 30 respondents selected from five barangays. The participants included SK Chairpersons, SK Secretaries, SK Treasurers with each barangay represented by its set of officials. These individuals were chosen because they are directly involved in SK operations and document management, making them the most suitable sources of information for the study.

The researchers used this group of respondents to support both quantitative and qualitative approaches. This allowed the collection of numerical data through structured responses, as well as deeper insights based on the participants' experiences and perspectives. By including key SK officials, the study ensured that the data gathered is relevant, reliable, and reflective of actual SK administrative practices.

### *D. Research Instrument*

This study used both quantitative and qualitative methods to gather data from the respondents. For the quantitative part, a structured questionnaire with checklist and rating-scale questions was created and distributed through Google Forms. This made it easier for the respondents to answer and for the researchers to quickly collect and organize the data. The questions mainly focused on evaluating SKActHub in terms of usability, efficiency, functionality, and overall performance.

To support the numerical data, the questionnaire also included open-ended questions. These allowed the respondents to freely share their thoughts, experiences, and suggestions while using the system. This gave the researchers a better understanding of what users felt about the system beyond just the ratings.

By combining both survey responses and written feedback, the study gathered both measurable results and real user insights. This helped provide a more complete and meaningful evaluation of SKActHub, especially in identifying its strengths and areas that still need improvement.

***E.Project Description***

SKActHub is a web-based system designed to enhance the management of Sangguniang Kabataan (SK) documents, activities, and membership information. The system enables SK officials to efficiently upload, organize, and archive important records such as activity proposals, accomplishment reports, and approval documents in a centralized digital repository. It also provides real-time monitoring of submitted documents and ongoing activities, allowing officials to track progress, identify delays, and ensure timely completion of tasks. In addition, SKActHub includes a membership management feature that allows administrators to maintain and update SK member records, including monitoring membership status such as active, inactive, or resigned members. To ensure data security and confidentiality, the system incorporates role-based access control, restricting access to sensitive information based on user roles and responsibilities. By transitioning from traditional paper-based processes to a structured digital workflow, SKActHub improves transparency, strengthens accountability, and enhances overall operational efficiency within SK offices.

**Technology Used**

The SKActHub system utilizes a combination of modern web technologies to ensure efficient performance, scalability, and user-friendly interaction. At the core of the system is MongoDB, a NoSQL document-oriented database that stores SK documents, member records, activity logs, and other system data in flexible JSON-like structures. Supporting the backend is Node.js, a runtime environment that executes server-side code, along with Express.js, a lightweight framework responsible for handling routing, API requests, and overall server-side logic. These technologies work together to process system operations and connect the database with the frontend interface.

On the client side, React.js is used to build dynamic and interactive user interfaces, enabling the creation of dashboards, document upload pages, and monitoring tools. To enhance the design and responsiveness of the system, Tailwind CSS is implemented as a utility-first UI framework, ensuring a clean and modern layout across devices. Communication between the frontend and backend is facilitated by Axios, which allows seamless HTTP requests and data exchange.

For data management and structure, Mongoose serves as an Object Data Modeling (ODM) library that simplifies database operations and enforces schema organization. Security is maintained through JSON Web Token (JWT), which provides a token-based authentication system to secure user access and manage sessions effectively. Additionally, Chart.js is used to present system data visually through charts and analytics, improving monitoring and decision-making.

To support development and collaboration, the project uses:

- Git and GitHub for version control, enabling developers to track changes and manage the source code efficiently
- Visual Studio Code as the primary development environment for coding, debugging, and maintaining the system

Overall, these technologies collectively ensure that the SKActHub system is robust, secure, and capable of delivering a responsive and efficient user experience.

**Statistical Treatment**

To ensure systematic and objective interpretation of the collected data, appropriate statistical tools were applied to both the pre-survey and post-survey questionnaire responses.

For the Pre-Survey Questionnaire, frequency distribution and percentage were used to summarize the respondents' answers. These statistical measures helped identify the prevailing document management practices, common challenges, retrieval efficiency, and readiness for digital system adoption among the selected Sangguniang Kabataan (SK) offices.

The formula used to compute the percentage is:

$$\text{Percentage} = \frac{f}{N} \times 100$$

where:

$f$ = frequency of responses

$N$ = total number of respondents

This method allowed the researchers to determine the proportion of respondents selecting each response option and to establish baseline data before system implementation.

For the Post-Survey Questionnaire, a five-point Likert Scale was utilized to measure respondents' level of agreement regarding the usability, effectiveness, security, and overall performance of SKActHub. The responses were analyzed using the *Weighted Mean* to determine the average rating for each statement and each evaluation category.

The formula for the Weighted Mean is:

$$\text{Weighted Mean} = \frac{\sum(f \times x)}{N}$$

where:

$f$ = frequency of each response

$x$ = numerical value assigned to each Likert scale option

$N$ = total number of respondents

The computed weighted mean was interpreted using the following scale:

**Mean Range  Verbal Interpretation**

**Table** 1 presents the mean ranges and their corresponding verbal interpretations used to analyze the respondents' level of agreement.

| | |
|---|---|
| 4.21 – 5.00 | Strongly Agree |
| 3.41 – 4.20 | Agree |
| 2.61 – 3.40 | Neutral |

| | |
|---|---|
| 1.81 – 2.60 | Disagree |
| 1.00 – 1.80 | Strongly Disagree |

In addition to the weighted mean, *Standard Deviation (SD)* was used to measure the variability or dispersion of responses. Standard deviation determines how close or spread out the responses are from the computed mean. A low standard deviation indicates that the responses are closely clustered around the mean, suggesting consistent evaluations among respondents. A high standard deviation indicates greater variability in responses.

The formula used for Standard Deviation is:

$$SD = \sqrt{\frac{\sum(x - \bar{x})^2}{N}}$$

where:

$x$ = individual response value

$\bar{x}$= mean

$N$ = total number of respondents

By computing both the weighted mean and standard deviation, the study ensures a more comprehensive analysis of the system evaluation results. The weighted mean determines the overall level of agreement, while the standard deviation provides insight into the consistency of respondents' perceptions. These statistical tools collectively support an objective and reliable assessment of SKActHub's effectiveness in improving document management, activity monitoring, and SK member administration.

### Funtional and Non-Functional Feature

The SKActHub system is designed to support the administrative and operational needs of the Sangguniang Kabataan by providing a set of well-defined functional and non-functional features. These features ensure that the system not only performs its intended tasks effectively but also maintains quality, security, and reliability in its overall performance.

The functional features focus on the core services and operations of the system. These are the specific actions that users can perform to manage SK activities and documents efficiently. In general, the system enables users to:

- Securely access the platform through user authentication using registered usernames and passwords
- Manage user accounts, including creating, updating, and assigning roles or permissions
- Upload important SK documents such as proposals, resolutions, and accomplishment reports
- Store documents in a centralized and secure digital repository for easy access
- Organize files into categories like proposals, reports, approvals, and financial records
- Search for documents quickly using filters such as title, date, or category
- Track the status of documents and activities (e.g., pending, approved, completed)
- Monitor ongoing SK activities and view real-time updates

- View and download documents as needed by authorized users
- Receive notifications regarding updates, approvals, or status changes
- Maintain user activity logs to ensure accountability and transparency
- Archive inactive documents securely, allowing retrieval when necessary while preventing accidental modification or deletion

On the other hand, the non-functional features describe the quality standards and system attributes that ensure a smooth and reliable user experience. These features do not focus on specific actions but rather on how well the system performs. The system is designed to:

- Provide fast performance, with pages and documents loading within a few seconds
- Ensure strong security through authentication, authorization, and protected data storage
- Maintain reliability by minimizing system downtime and ensuring consistent operation
- Offer high usability with a user-friendly interface that is easy for SK officials to navigate
- Support scalability, allowing the system to handle more users and data without slowing down
- Maintain compatibility across modern web browsers such as Chrome, Edge, and Firefox

- Allow easy maintenance, making future updates and improvements manageable

- Perform regular data backups to prevent loss of important documents

Overall, these functional and non-functional features work together to make the SKActHub system efficient, secure, and dependable for managing SK-related activities and records.

# CHAPTER IV

# PRESENTATION AND ANALYSIS OF DATA

This chapter presents the data gathered during the testing and evaluation of the developed system. The results are organized and presented using tables and descriptive analysis to clearly show the performance and effectiveness of the system. The data collected from respondents and system testing are analyzed to determine whether the objectives of the study were achieved. The findings in this chapter serve as the basis for interpreting the results and assessing the overall functionality and usability of the developed system

***Level of Aggreement***

Table 3 presents the summary of the pre-survey results based on the responses of thirteen (13) participants out of the target twenty-five (25) respondents. The data were analyzed using frequency and percentage to determine the current practices in document management, monitoring, communication, and system readiness among SK officials. The results indicate that most respondents still rely on manual or partially digital processes, which affect efficiency, accessibility, and overall workflow.

**Table 3. Pre-Survey Summary of Current Practices (N = 13)**

| Indicators | Most Common Response | Percentage |
|---|---|---|
| Document Storage | Paper-based with some digital | 90% |
| Difficulty in Locating Documents | Sometimes | 55% |
| Document Retrieval Time | 5–15 minutes | 52% |
| Lost or Misplaced Documents | A few times | 68% |

| Indicators | Most Common Response | Percentage |
|---|---|---|
| Membership Record Management | Manual Logbook | 55% |

**Table 3.1 Monitoring and Reporting (N = 13)**

| Indicators | Most Common Response | Percentage |
|---|---|---|
| Activity Tracking Method | Printed Reports | 60% |
| Ease of Generating Reports | Neutral | 48% |
| Centralized Monitoring System | Yes | 65% |
| Delays in Submission | Sometimes | 53% |

**Table 3.2 Communication and Accessibility (N = 13)**

| Indicators | Most Common Response | Percentage |
|---|---|---|
| Communication Method | Messenger/Chat Apps | 68% |
| Centralized Messaging System | Yes | 70% |
| Accessibility of Documents | Moderately Accessible | 45% |

**Table 3.3 System Needs and Readiness (N = 13)**

| Indicators | Most Common Response | Percentage |
|---|---|---|
| Need for Digital System | Strongly Agree | 75% |
| Willingness to Adopt System | Yes | 80% |

**Table 3: Document Management**

| Indicator | % | f |
|---|---|---|
| Document Storage | 90% | 12 |
| Difficulty in Locating Documents | 55% | 7 |
| Document Retrieval Time | 52% | 7 |
| Lost or Misplaced Documents | 68% | 9 |
| Membership Record Management | 55% | 7 |
| **TOTAL** | | **29** |

**Table 3.1: Monitoring and Reporting**

| Indicator | % | f |
|---|---|---|
| Activity Tracking Method | 60% | 8 |
| Ease of Generating Reports | 48% | 6 |
| Centralized Monitoring System | 65% | 8 |
| Delays in Submission | 53% | 7 |
| **TOTAL** | | **29** |

**Table 3.2: Communication and Accessibility**

| Indicator | % | f |
|---|---|---|
| Communication Method | 68% | 9` |

| Indicator | % | f |
|---|---|---|
| Centralized Messaging System | 70% | 9 |
| Accessibility of Documents | 45% | 6 |
| **TOTAL** | | **24** |

**Table 3.3: System Needs and Readiness**

| Indicator | % | f |
|---|---|---|
| Need for Digital System | 75% | 10 |
| Willingness to Adopt System | 80% | 10 |
| **TOTAL** | | **20** |

**Table 3.4** Combined Summary of Pre-Survey Results **(N = 13)**

| Category | No. of Indicators | Total f | Total Responses | Percentage |
|---|---|---|---|---|
| Document Management | 5 | 42 | 65 | 65% |
| Monitoring and Reporting | 4 | 29 | 52 | 56% |
| Communication and Accessibility | 3 | 24 | 39 | 62% |
| System Needs and Readiness | 2 | 20 | 26 | 77% |
| TOTAL | 14 | 115 | 182 | 63% |

The data in Table 3 and its subsections show that most SK offices still depend on manual and semi-digital systems, which results in challenges such as time-consuming document retrieval, occasional loss of records, and limited accessibility. In terms of monitoring and reporting, the lack of a centralized system contributes to delays and inefficiencies in tracking activities and generating reports. These findings are further supported by the overall results, which indicate a total weighted distribution of 14, 115, and 182, with an overall percentage of 63%. This suggests that while some level of digital practice exists, many processes are still not fully optimized or systematized, reinforcing the need for an integrated system like SKActHub to improve efficiency, accuracy, and accessibility in SK operations.

***Level of Aggreement***

Table 4 presents the post-survey evaluation of the SKActHub system based on the responses of sixteen (16) participants out of the expected twenty-five (25), yielding a response rate of 64%. The data were analyzed using the weighted mean and interpreted using the given descriptive rating scale.

The results show that respondents generally have a positive evaluation of the system across its major features. *Overall System Effectiveness* obtained the highest weighted mean of 4.25 (Strongly Agree), indicating that the system is perceived as highly effective. In contrast, *Activity Monitoring and Tracking* received the lowest weighted mean of 3.31 (Neutral), suggesting that respondents have mixed opinions regarding this feature.

Meanwhile, *System Usability and User Experience* (4.25) and *Document Management and Organization* (4.25) are both interpreted as Strongly Agree, reflecting high user satisfaction. On the other hand, *SK Member Management* (3.78) and *Security* *and Access Control* (4.10) are interpreted as Agree, indicating that these features are acceptable but may still require improvement.

Overall, the computed general weighted mean of 3.92 (Agree) shows that the SKActHub system is generally functional and acceptable. However, certain areas particularly activity monitoring may still need enhancement to further improve the overall user experience.

**Table 4. Post-Survey Evaluation of SKActHub (N = 16)**

**Section A: System Usability and User Experience**

| Rating | Freq |
|---|---|
| 5 | 42 |
| 4 | 42 |
| 3 | 6 |
| 2 | 6 |
| 1 | 0 |
| WM = 4.25 | Total = 96(Strongly Agree) |

**Section B: Document Management and Organization**

| Rating | Freq |
|---|---|
| 5 | 42 |
| 4 | 42 |
| 3 | 6 |
| 2 | 6 |

| Rating | Freq |
|---|---|
| 1 | 0 |
| WM = 4.25 | Total = 96(Strongly Agree) |

**Section C: Activity Monitoring and Tracking**

| **Rating** | **Freq** |
|---|---|
| 5 | 10 |
| 4 | 25 |
| 3 | 30 |
| 2 | 10 |
| 1 | 5 |
| WM = 3.31 | Total = 80 (Neutral) |

**Section D: SK Member Management**

| **Rating** | **Freq** |
|---|---|
| 5 | 12 |
| 4 | 21 |
| 3 | 12 |
| 2 | 3 |
| 1 | 0 |
| WM =3.87 | Total = 48 (Agree) |

**Section E: Security and Access Control**

| Rating | Freq |
|---|---|
| 5 | 18 |
| 4 | 21 |
| 3 | 6 |
| 2 | 3 |
| 1 | 0 |
| WM = 4.10 | Total = 48(Agree) |

**Section F: Overall System Effectiveness**

| Rating | Freq |
|---|---|
| 5 | 3 |
| 4 | 7 |
| 3 | 5 |
| 2 | 1 |
| 1 | 0 |
| WM = 3.75 | Total = 16(Agree) |

| **Indicators** | **WM** | **DR** |
|---|---|---|
| 1. System Usability and User Experience | 4.25 | Strongly Agree |
| 2. Document Management and Organization | 4.25 | Strongly Agree |
| 3. Activity Monitoring and Tracking | 3.31 | Neutral |
| 4. SK Member Management | 3.87 | Agree |
| 5. Security and Access Control | 4.10 | Agree |
| 6. Overall System Effectiveness | 3.75 | Agree |
| Total Weighted Mean | 23.53 | |
| General Weighted Mean | 3.92 | Agree |

**Testing Results:**

**Table 4**

**Login Test**

| Test Case ID | Test Scenario | Test Steps | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|
| TC001 | Login with valid credentials | 1. Open login page2. Enter valid email & password3. Click login | User should be logged in and redirected to dashboard | User redirected successfully | ☑ Pass |
| TC002 | Login with invalid password | 1. Enter correct email2. Enter wrong password3. Click login | Error message should appear | Error displayed: "Invalid password" | ☑ Pass |
| TC003 | Empty input fields | 1. Leave email & password empty2. Click login | Validation error should show | "Fields required" shown | ☑ Pass |

***Figure 7 : Login Test***

**Table 5**

**Add Event**

| Test Case ID | Test Scenario | Preconditions | Test Steps | Expected Result | Status |
|---|---|---|---|---|---|
| TC-E01 | Login with valid admin credentials | System is running | 1. Open login page2. Enter username & password3. Click Sign In | User redirected to Events page | Pass |
| TC-E02 | Open Create Event modal | User is logged in | 1. Click "Create Event" button | Modal should appear | Pass |
| Test Case ID | Test Scenario | Preconditions | Test Steps | Expected Result | Status |
| TC-E03 | Empty form submission validation | Modal is open | 1. Click Create Event without input | Browser validation message for required fields (e.g., title) | Pass |
| Test Case ID | Test Scenario | Preconditions | Test Steps | Expected Result | Status |
| TC-E04 | Input event title and description | Modal is open | 1. Enter title2. Enter description | Inputs should accept values | Pass |
| TC-E05 | Input start and end date | Modal is open | 1. Enter start date2. Enter end date | Dates should be accepted | Pass |
| Test Case ID | Test Scenario | Preconditions | Test Steps | Expected Result | Status |
| TC-E06 | Select "Specific" visibility | Modal is open | 1. Click "Specific" radio button | Barangay dropdown should appear | Pass |
| TC-E07 | Barangay dropdown loads options | Dropdown visible | 1. Wait for options2. Check options count | Options should be loaded | Pass |

| | | | | | |
|---|---|---|---|---|---|
| TC-E08 | Select barangay | Dropdown loaded | 1. Select barangay option | Barangay should be selected | Pass |
| **Test Case ID** | **Test Scenario** | **Preconditions** | **Test Steps** | **Expected Result** | **Status** |
| TC-E09 | Submit valid event form | All fields filled | 1. Fill form correctly2. Click Create Event | Success or response message should appear | Pass |
| TC-E10 | Display submission feedback | Form submitted | 1. Observe message area | System should show success or error message | Pass |
| **Test Case ID** | **Test Scenario** | **Preconditions** | **Test Steps** | **Expected Result** | **Status** |
| TC-E11 | Close modal using Cancel button | Modal is open | 1. Click Cancel | Modal should close | Pass |

***Figure 8 : Add Event Test***

**Edit Official**

**Table 6**

| Test Case ID | Test Scenario | Preconditions | Test Steps | Expected Result | Status |
|---|---|---|---|---|---|
| TC-EO01 | Login with valid admin credentials | System is running | 1. Open login page2. Enter username & password3. Click Sign In | User redirected to SK Officials page | Pass |
| TC-EO02 | Open Edit Official modal | User is logged in | 1. Click "More" button2. Click "Edit" | Edit modal should appear | Pass |

| Test Case ID | Test Scenario | Preconditions | Test Steps | Expected Result | Status |
|---|---|---|---|---|---|
| TC-EO03 | First name required validation | Modal is open | 1. Clear first name2. Click Save Changes | Browser validation message should appear | Pass |
| TC-EO04 | Last name required validation | Modal is open | 1. Clear last name2. Click Save Changes | Validation message should appear | Pass |
| TC-EO05 | Position required validation | Modal is open | 1. Clear/select empty position2. Submit form | Validation message should appear | Pass |
| TC-EO06 | Barangay required validation | Modal is open | 1. Clear/select empty barangay2. Submit form | Validation message should appear | Pass |
| **Test Case ID** | **Test Scenario** | **Preconditions** | **Test Steps** | **Expected Result** | **Status** |
| TC-EO07 | Update basic fields | Modal is open | 1. Enter new first name2. Enter new last name3. Select position4. Select barangay | Fields should update successfully | Pass |
| **Test Case ID** | **Test Scenario** | **Preconditions** | **Test Steps** | **Expected Result** | **Status** |

| | | | | | |
|---|---|---|---|---|---|
| TC-EO08 | Open password reset fields | Modal is open | 1. Click "Reset Password" | Password fields should appear | Pass |
| TC-EO09 | Password mismatch validation | Password fields visible | 1. Enter password2. Enter different confirm password3. Submit | Error "Passwords do not match" should appear | Pass |
| TC-EO10 | Correct password confirmation | Password mismatch fixed | 1. Match confirm password with password | Error should disappear | Pass |
| **Test Case ID** | **Test Scenario** | **Preconditions** | **Test Steps** | **Expected Result** | **Status** |
| TC-EO12 | Cancel button closes modal | Modal is open | 1. Click Cancel | Modal should close | Pass |

***Figure 9 : Edit Official Test***

**Create User Official**

**Table 7**

| Test Case ID | Test Scenario | Preconditions | Test Steps | Expected Result | Status |
|---|---|---|---|---|---|
| TC-O01 | Login with valid admin credentials | System is running | 1. Open login page2. Enter username & password3. Click Sign In | User redirected to SK Officials page | Pass |
| TC-O02 | Open Create SK Official modal | User is logged in | 1. Click "Add Official" button | Modal should appear | Pass |

| | | | | | |
|---|---|---|---|---|---|
| TC-O03 | Empty form validation | Modal is open | 1. Click Create without input | Validation errors should appear (e.g., First name required) | Pass |
| TC-O04 | Input first and last name | Modal is open | 1. Enter first name2. Enter last name | Inputs should accept values | Pass |
| TC-O05 | Username auto-generation | Names entered | 1. Enter first & last name2. Check username field | Username should be auto-generated | Pass |
| TC-O06 | Dropdown selection (Position & Barangay) | Modal is open | 1. Select position2. Select barangay | Dropdown values should be selectable | Pass |
| TC-O07 | Password mismatch validation | Modal is filled | 1. Enter password2. Enter different confirm password3. Submit | Error message "Passwords do not match" should appear | Pass |
| TC-O08 | Correct password confirmation | Password exists | 1. Match confirm password with password | Error should disappear | Pass |
| TC-O09 | Password strength indicator | Password field filled | 1. Enter password2. Observe UI | Password strength indicator should be visible | Pass |
| TC-O10 | Toggle password visibility | Password field filled | 1. Click toggle button | Password visibility should change | Pass |
| TC-O11 | Submit valid form | All fields valid | 1. Fill all fields correctly2. Click Create Official | Modal should close (success) | Pass |
| TC-O12 | Prevent submission if errors exist | Form has validation errors | 1. Try submitting invalid form | Form should not submit | Pass |

| | | | | | |
|---|---|---|---|---|---|
| TC-O13 | API error handling (No token) | User logged out (token removed) | 1. Remove token2. Refresh page | System should show error / unauthorized behavior | Pass |
| TC-O14 | API failure feedback | Submission fails | 1. Submit form2. Simulate API error | Error message/banner should appear | Pass |

***Figure 10 : Create User Official***

**Folder Creation**

**Table 8**

| **Test Case ID** | **Test Scenario** | **Preconditions** | **Test Steps** | **Expected Result** | **Status** |
|---|---|---|---|---|---|
| TC-F01 | Cancel button closes Create Folder modal | User is logged in and inside Barangay Storage | 1. Click "New Folder"2. Click "Cancel" | Modal should close | Pass |
| TC-F02 | X button closes Create Folder modal | Same as above | 1. Open Create Folder modal2. Click "X" button | Modal should close | Pass |
| TC-F03 | Create button disabled when input is empty | Modal is open | 1. Do not input folder name2. Check Create button | Button should be disabled | Pass |
| TC-F04 | Create folder successfully | Modal is open | 1. Enter folder name2. Click Create | Folder should appear in list | Pass |

| | | | | | |
|---|---|---|---|---|---|
| TC-F05 | Create folder using Enter key | Modal is open | 1. Enter folder name2. Press Enter | Folder should be created | Pass |
| TC-F06 | Duplicate folder name is rejected | Folder already exists | 1. Enter existing folder name2. Click Create | Error message should appear ("already exists") | Pass |
| TC-F07 | Open folder view modal | Folder exists | 1. Click folder card | Folder modal should open with search bar | Pass |
| TC-F08 | Search inside folder (empty state) | Folder view modal is open | 1. Type random text in search2. Observe result | "No documents found" message appears | Pass |
| TC-F09 | Close folder view modal using Close button | Folder modal is open | 1. Click "Close" | Modal should close | Pass |
| **Test Case ID** | **Test Scenario** | **Preconditions** | **Test Steps** | **Expected Result** | **Status** |
| TC-F11 | Add Document button opens compose modal | Folder exists | 1. Click "+" button | Compose modal should open | Pass |
| TC-F12 | Compose submit disabled when fields empty | Compose modal open | 1. Leave fields empty2. Check button | Button should be disabled | Pass |

| | | | | | |
|---|---|---|---|---|---|
| TC-F13 | Cancel button closes compose modal | Compose modal open | 1. Click Cancel | Modal should close | Pass |
| TC-F14 | X button closes compose modal | Compose modal open | 1. Click X | Modal should close | Pass |
| **Test Case ID** | **Test Scenario** | **Preconditions** | **Test Steps** | **Expected Result** | **Status** |
| TC-F15 | Set folder status to Ongoing | Folder exists | 1. Click "Set Ongoing"2. Confirm | Status should change to Ongoing | Pass |
| TC-F16 | Cancel status change | Status modal open | 1. Click "Set Completed"2. Click Cancel | Status should remain unchanged | Pass |
| TC-F17 | Set folder status to Completed | Folder exists | 1. Click "Set Completed"2. Confirm | Status should change to Completed | Pass |
| **Test Case ID** | **Test Scenario** | **Preconditions** | **Test Steps** | **Expected Result** | **Status** |
| TC-F18 | Cancel delete folder | Folder exists | 1. Click Delete2. Click Cancel | Folder should NOT be deleted | Pass |
| TC-F19 | Confirm delete folder | Folder exists | 1. Click Delete2. Click Confirm | Folder should be removed from list | Pass |
| TC-F20 | Cleanup test folder | Test folder exists | 1. Delete test folder | Folder should be removed | Pass |

**User Acceptance Testing (UAT)**

Before fully implementing the system, User Acceptance Testing (UAT) was carried out to make sure that it works well for its intended users. This step allowed actual SK officials to try the system and see if it fits their needs and daily tasks.

During this phase, selected users interacted with the system in a real-world setting. They tested its features, explored its functions, and shared their experiences while using it. This helped the researchers understand how easy the system is to use, how well it performs, and whether it meets user expectations.

The main goal of UAT is to confirm that the system is ready for use. It also provides an opportunity to gather feedback, identify minor issues, and make necessary improvements. Through this process, the researchers were able to ensure that the system is not only functional, but also practical, reliable, and user-friendly for SK officials.

**A. Admin Module – SKActHub**

**Table 9**

| **Test Scenario** | **Objective** | **Process** | **Expected Result** | **Actual Result** | **Status** |
|---|---|---|---|---|---|
| 1. Admin Login | To verify admin access | Enter credentials and click login | Redirect to dashboard | Successfully logged in | Passed |
| 2. View Pending Messages | To view submitted documents | Open pending messages | Display message details | Displayed correctly | Passed |
| 3. Approve Documents | To approve submissions | Click approve | Document stored properly | Successfully approved | Passed |
| 4. Reject Documents | To reject invalid documents | Click reject | Document not stored | Successfully rejected | Passed |
| 5. Create SK Official | To create accounts | Fill form and submit | Account created | Successfully created | Passed |
| 6. Create Event Schedule | To create system events | Input event details and submit | Event saved | Event created successfully | Passed |
| 7. User Logs Monitoring | To track system activity | Filter logs by name/type/date | Logs displayed | Logs shown correctly | Passed |

## B. SK Official Module – SKActHub

**Table 10**

| **Test Scenario** | **Objective** | **Process** | **Expected Result** | **Actual Result** | **Status** |
|---|---|---|---|---|---|
| 1. Official Login | To verify official access | Enter credentials | Redirect to dashboard | Logged in successfully | Passed |
| 2. Inbox (Receive & Sent)** | To view communication records | Open inbox → check received & sent tabs | Messages displayed correctly | Messages shown properly | Passed |
| 3. Create Event | To allow official to create events | Input title, description, start/end date & visibility | Event saved and displayed | Event created successfully | Passed |
| 4. Barangay Folder Management | To manage document folders | Create or delete folder | Folder created/deleted | Successfully managed folders | Passed |
| 5. Submit Document for Approval | To send documents to admin | Enter subject, details, attach file, then submit | Document sent to admin for approval | Document successfully submitted | Passed |
| 6. SK Personnel Management | To manage SK member status | Input surname, first name, middle initial, age, set status (active/inactive) | Member record saved and updated | Member status updated successfully | Passed |
| 7. View & Update Profile | To edit official profile | Update personal details and save | Changes reflected | Successfully updated | Passed |
| 8. Logout | To securely exit system | Click logout | Redirect to login | Logged out successfully | Passed |
| 9. Navigation & Usability | To evaluate ease of use | Navigate modules | Smooth navigation | User-friendly interface | Passed |

# CHAPTER V

# SUMMARY FINDINGS, CONCLUSION AND RECOMMENDATIONS

## Summary

This study focused on the development of SKActHub, a web-based system designed to improve document management, activity monitoring, and record organization within the Sangguniang Kabataan (SK). The researchers identified that many SK offices still rely on manual processes, which often result in misplaced documents, delays in approval, and difficulty in tracking records. To better understand these challenges, data were gathered through interviews, surveys, and observation involving SK officials such as the Chairman, Secretary, Treasurer, and other SK Kagawad.

Based on the findings, the researchers developed SKActHub as a centralized system with features such as document uploading with approval workflow, role-based access control, event management, and notifications. The system was built using the MERN stack and evaluated through User Acceptance Testing, where users found it functional, reliable, and easy to use. Overall, the system proved to be an effective solution that improves efficiency, enhances transparency, and supports better decision-making within the SK.

## Findings

Based on the data gathered and the evaluation of the system, several important findings were identified. SK officials currently experience difficulties in managing documents due to their reliance on manual and paper-based processes. As a result, documents are often misplaced, duplicated, or hard to retrieve, which affects overall efficiency. In addition, monitoring activities and tracking records take a significant amount of time and can sometimes be inconsistent, making it harder for officials to stay organized.

With the implementation of SKActHub, these challenges were significantly reduced. The system improved document organization by providing centralized digital storage, making it easier to manage and access files. The approval-based workflow added an extra layer of control by ensuring that only verified documents are stored, while role-based access control enhanced security by limiting user access based on their responsibilities. The system also performed well in terms of speed, especially in document retrieval and page loading, although document uploading takes slightly longer due to the approval process. Overall, users found the system easy to use, reliable, and helpful in completing their tasks more efficiently.

**Conclusions**

Based on the findings of the study, it can be concluded that the SKActHub system effectively addresses the common problems experienced in traditional document management and activity monitoring within the SK. The shift from manual, paper-based processes to a digital system provided a more organized, efficient, and secure way of handling records. Tasks that were previously time-consuming, such as searching for documents and tracking activities, became faster and more manageable through the system.

The use of a centralized digital platform significantly reduced the risk of lost, duplicated, or misplaced documents, while also making information more accessible to authorized users. The approval-based workflow played an important role in ensuring that all documents are properly reviewed before being stored, which helps maintain accuracy and accountability. At the same time, the implementation of role-based access control strengthened system security by ensuring that users can only access data relevant to their responsibilities.

Overall, SKActHub proved to be a practical and reliable solution that improves daily operations within the SK. It not only enhances efficiency and organization but also promotes transparency and better decision-making among officials. The positive feedback from users during testing shows that the system is easy to use and meets their needs, making it suitable for real-world implementation and future adoption in other barangays.

**Recommendations**

Based on the findings and conclusions of the study, the following recommendations are proposed:

***For the Sangguniang Kabataan (SK) Officials:*** It is recommended that they fully adopt and consistently utilize the SKActHub system in their daily operations to enhance efficiency in document management, activity monitoring, and record organization.

***For the System Users (SK Chairperson, Secretary, and Treasurer):*** Proper training and orientation programs should be conducted to ensure that users are equipped with the necessary knowledge and skills to effectively operate the system.

***For Future System Developers:*** It is recommended to further enhance the system by incorporating additional features such as mobile compatibility, automated approval workflows, and integration with other government information systems.

***For System Administrators:*** Regular system maintenance and updates should be implemented to ensure continuous functionality, data security, and overall system reliability.

***For Stakeholders and Users:*** Continuous collection and evaluation of user feedback are encouraged to support ongoing system improvement and ensure that the platform remains responsive to user needs.

***For Future Researchers:*** It is recommended to expand the scope of the study by implementing the system in other barangays or local government units and by exploring the integration of more advanced technologies to further improve digital governance.

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(Supports usability evaluation in your system testing)

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(If you used survey evaluation—very strong citation)

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(Strong support for your **Security and Access Control** feature)

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## Appendices A:

## System Features:

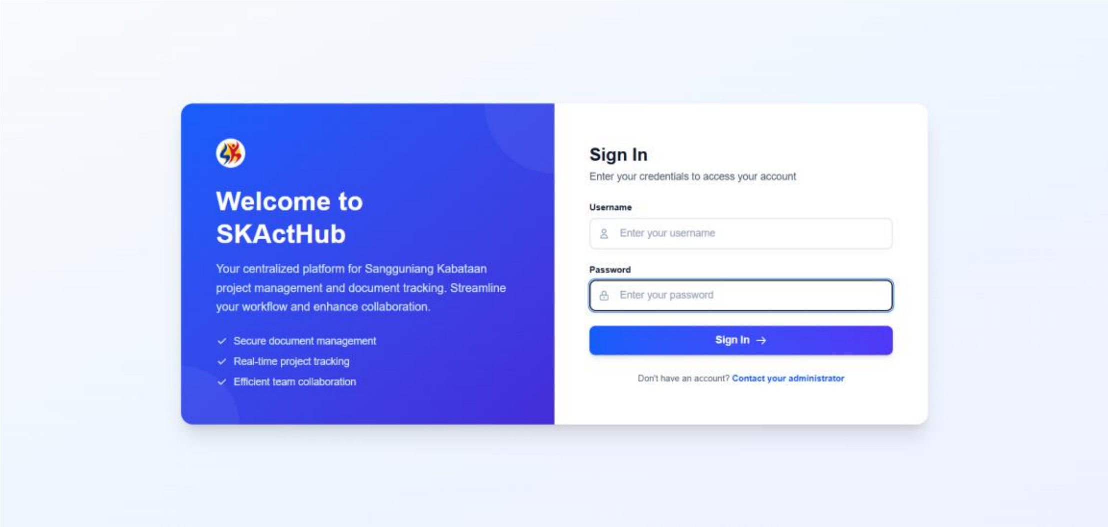



*Figure 8 : Login Of SkAct-Hub*

## *Admin Panel:*

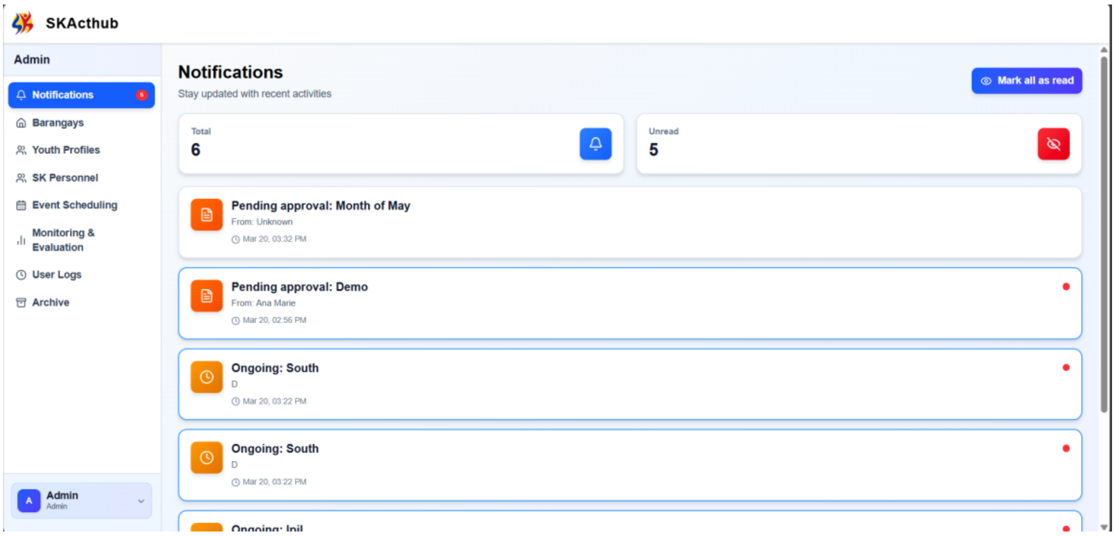



*Figure 9 : Notification Of SkAct-Hub*

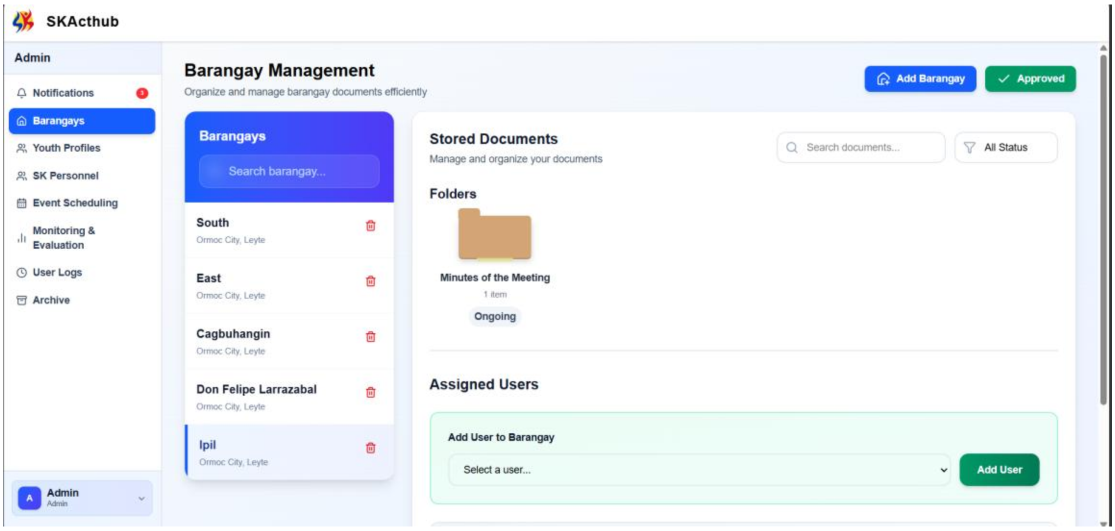



*Figure 10 : Barangay Management Of SkAct-Hub*

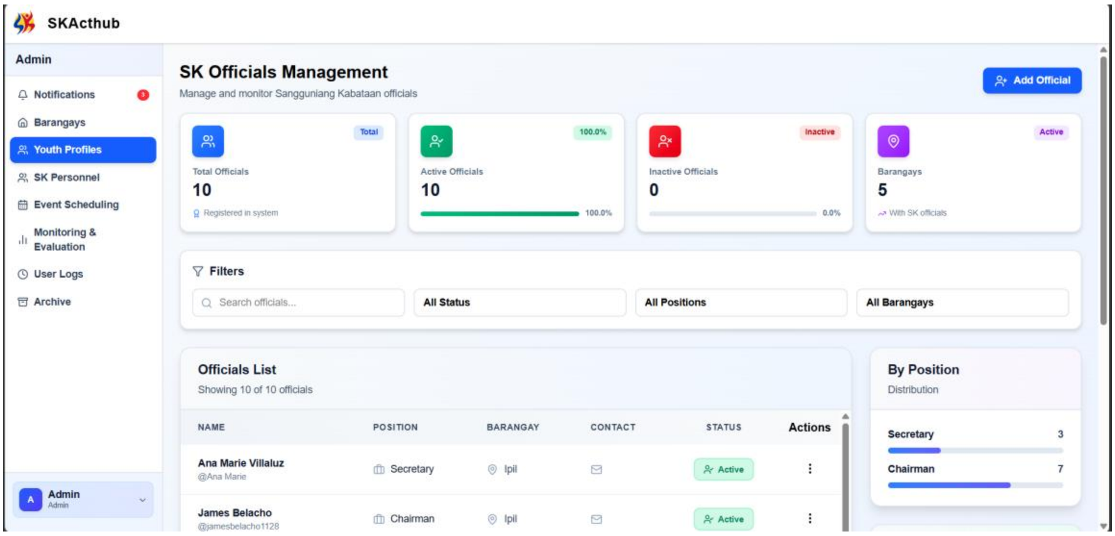



*Figure 11 : User Management Of SkAct-Hub*

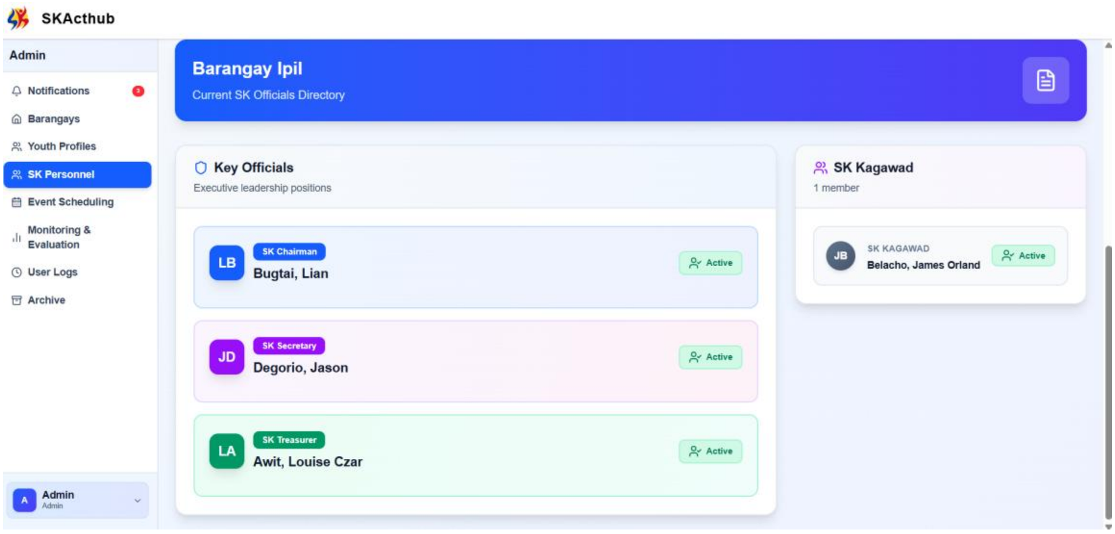



*Figure 12: Sk Official Status Of SkAct-Hub*

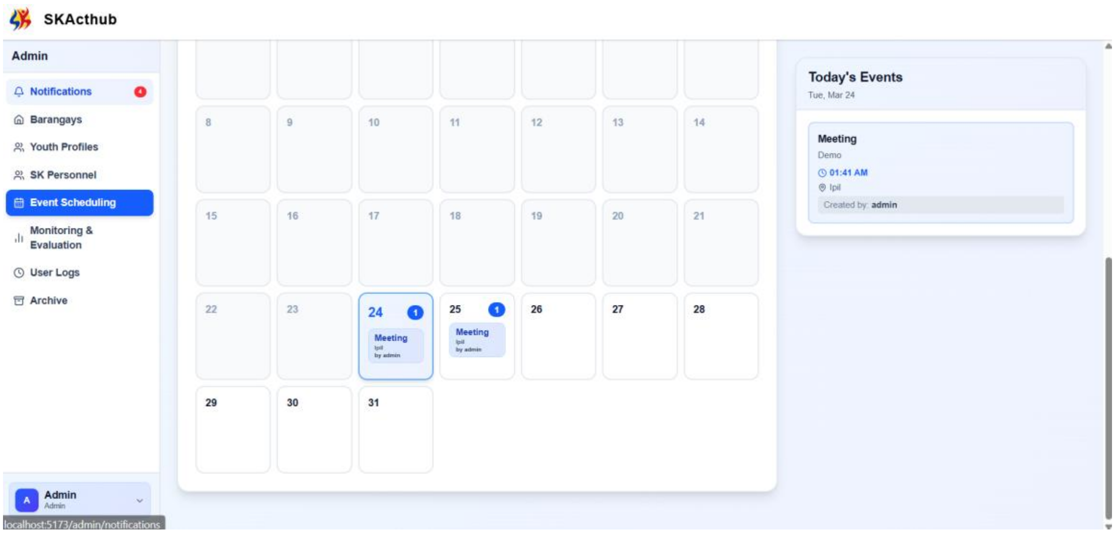



*Figure 13 : Event Scheduling Calendar Of SkAct-Hub*

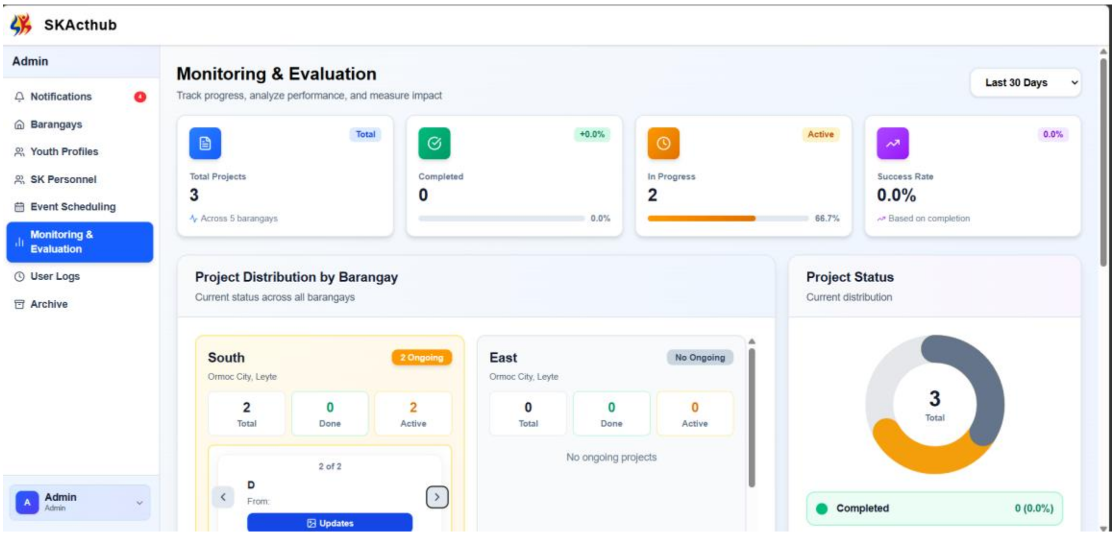



*Figure 14 : Monitoring & Evaluation Of SkAct-Hub*

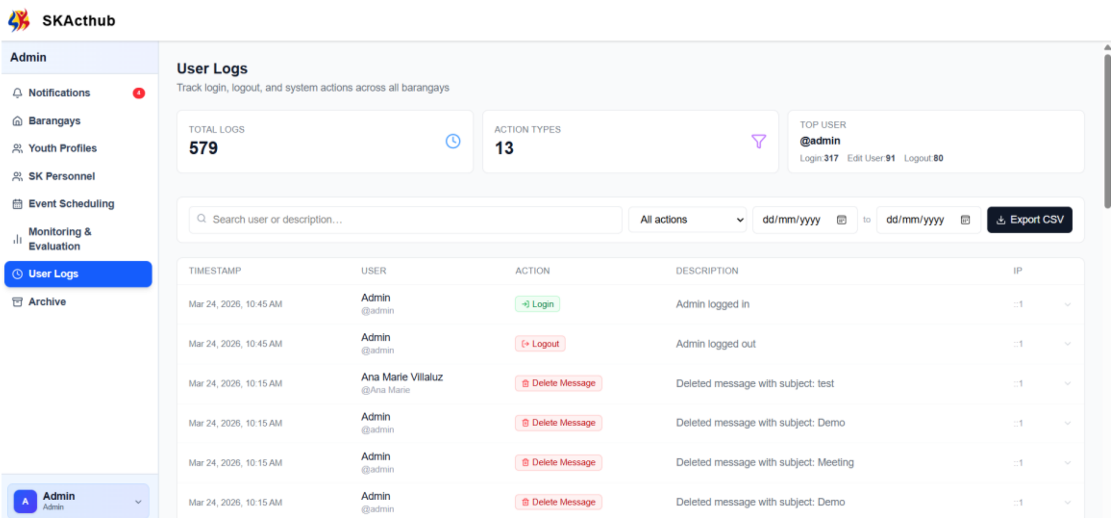



*Figure 15: User Logs Of SkAct-Hub*

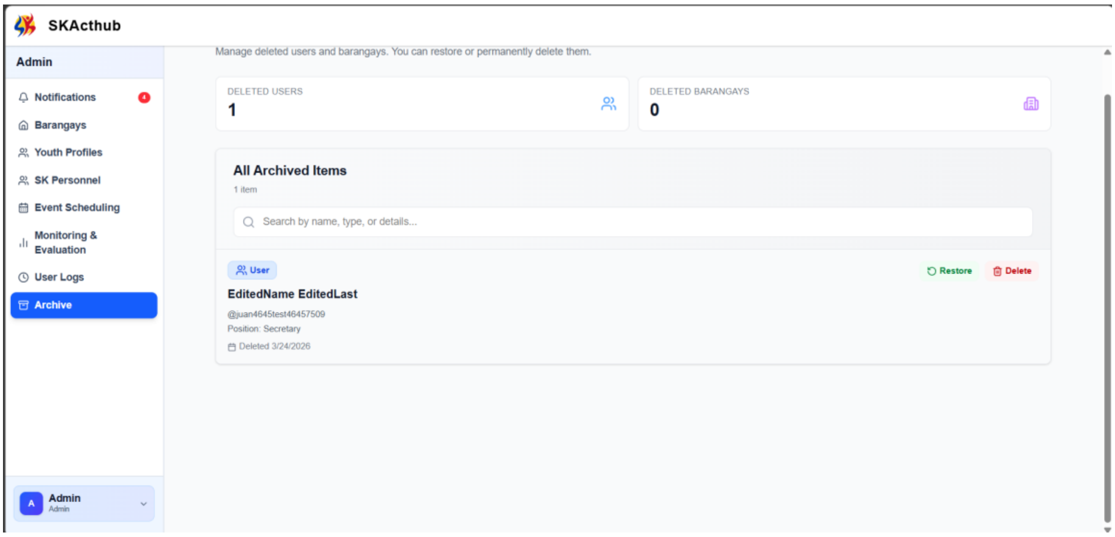



*Figure 16 : Archive Page Of SkAct-Hub*

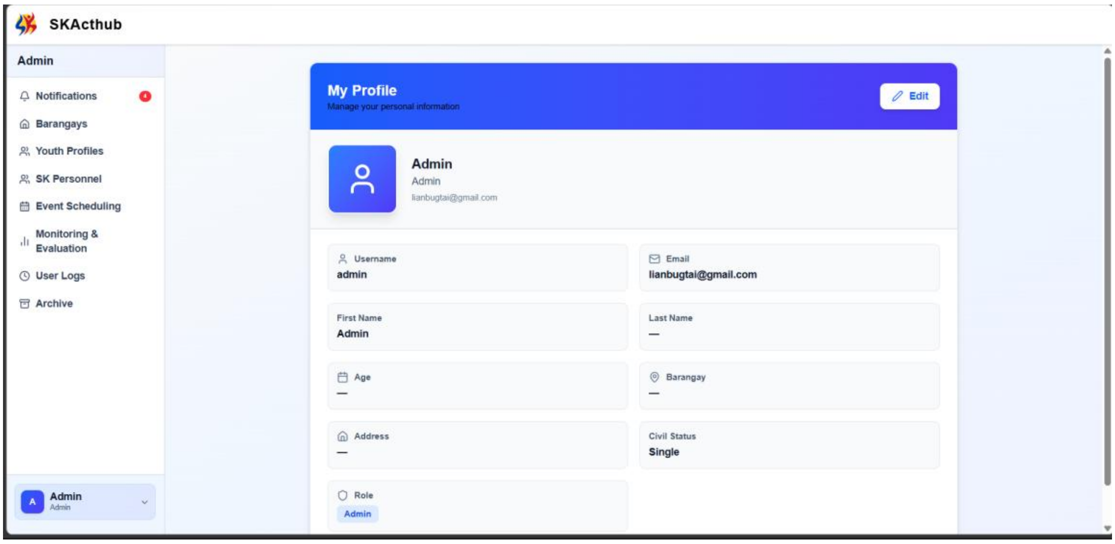



*Figure 17 : View Profile Of SkAct-Hub*

**Official Panel :**

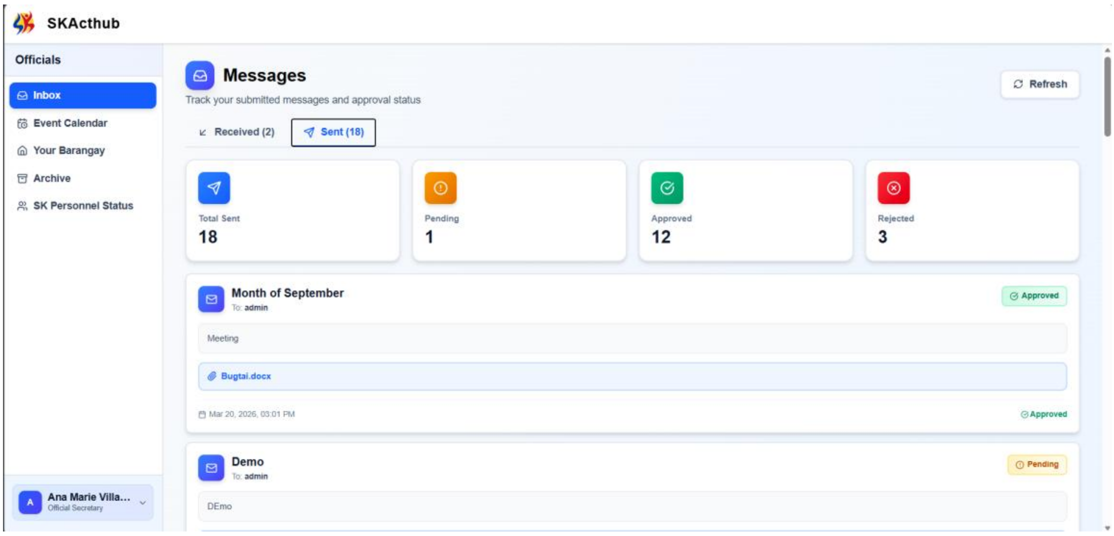



*Figure 18 : Inbox of the Official user in SkAct-Hub*

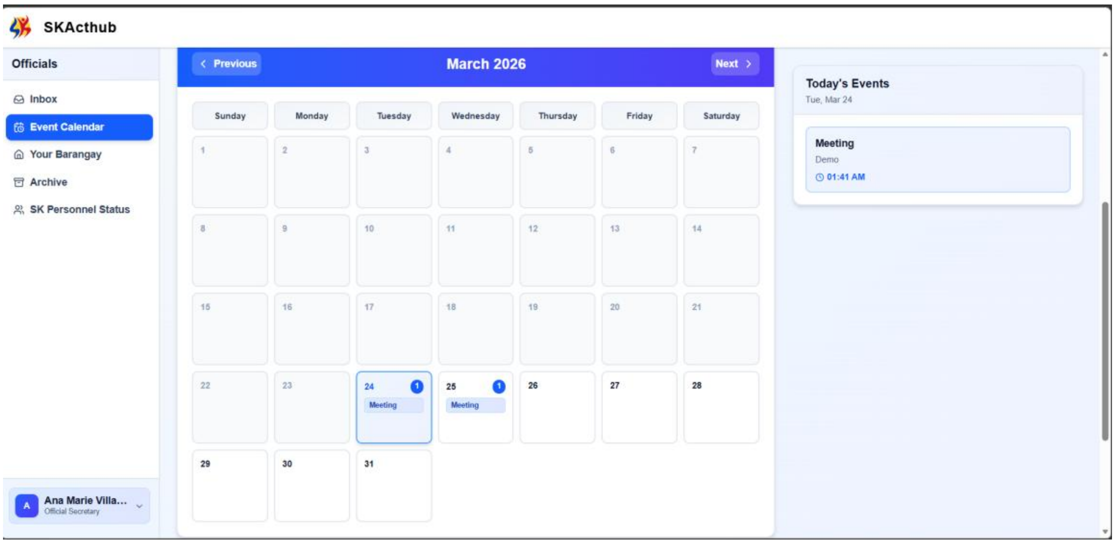



*Figure 19 : Event Scheduling of the Official User in SkAct-Hub*

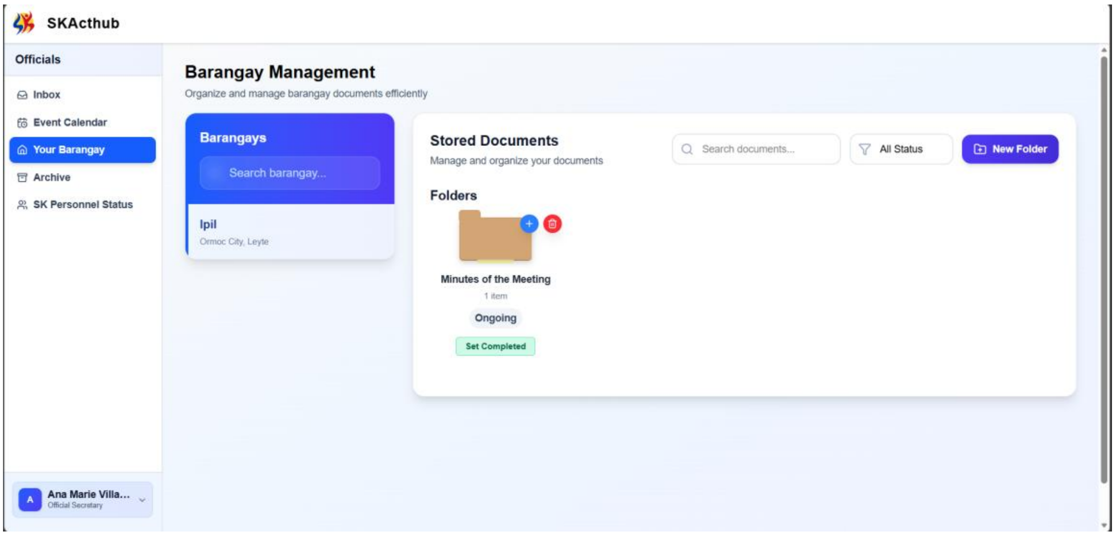



*Figure 20 : Stored Document of the Official User in SkAct-Hub*

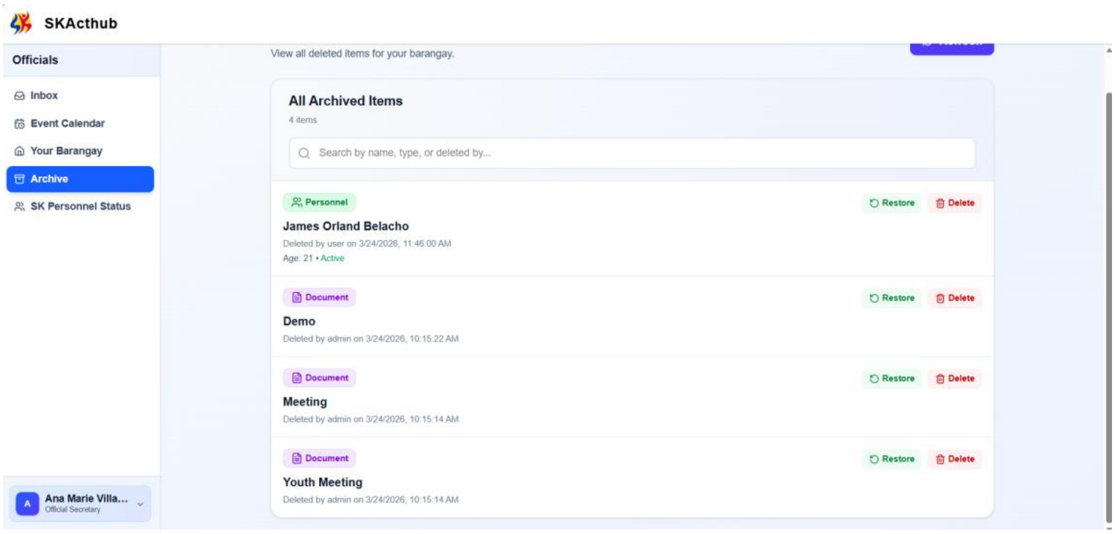



*Figure 19 : Archive Page of the Official User in SkAct-Hub*

**POST SURVEY QUESTIONAIRES**

**Name of Respondent: ___________________________**
**Barangay: ___________________________**

**Title:** SKActHub: A Project Management for Monitoring and Tracking SK Activities to Strengthen SK Governance for the Ormoc City Youth Development Office

**Instructions:**

Please rate the following statements based on your experience using **SKActHub**.

**Scale:**
1 – Strongly Disagree
2 – Disagree
3 – Neutral
4 – Agree
5 – Strongly Agree

Please check (✓) one answer for each statement.

**Section A: System Usability and User Experience**

**1. I can easily log in and access SKActHub.**
☐ 1 – Strongly Disagree
☐ 2 – Disagree
☐ 3 – Neutral
☐ 4 – Agree
☐ 5 – Strongly Agree

**2. The system interface is clear and organized.**
☐ 1 – Strongly Disagree
☐ 2 – Disagree
☐ 3 – Neutral
☐ 4 – Agree
☐ 5 – Strongly Agree

**3. I can easily navigate between different system modules.**
☐ 1 – Strongly Disagree
☐ 2 – Disagree
☐ 3 – Neutral
☐ 4 – Agree
☐ 5 – Strongly Agree

**4. Uploading and categorizing new documents is easy.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**5. Searching for specific documents is simple and efficient.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**6. Overall, I am satisfied with the usability of SKActHub.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**Section B: Document Management and Organization**

**7. SKActHub effectively organizes SK documents (proposals, reports, approvals).**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**8. Document retrieval is faster than the previous paper-based system.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**9. The system reliably prevents document loss or misplacement.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**10. I am satisfied with the document storage and archiving process in SKActHub.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**11. The system allows me to manage multiple types of documents simultaneously.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**12. I rarely encounter errors when uploading or retrieving documents.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**Section C: Activity Monitoring and Workflow Tracking**

**13. I can easily track the status of submitted SK activities or proposals.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**14. Notifications and alerts about activity updates are accurate and timely.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**15. SKActHub makes monitoring ongoing SK projects simple and efficient.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**16. The system effectively summarizes completed and pending tasks.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**17. Activity tracking dashboards are clear and informative for decision-making.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**Section D: SK Member Management**

**18. Viewing and updating SK member information is easy.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**19. I can effectively track membership status (active, inactive, resigned).**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**20. SKActHub keeps SK member records accurate and up-to-date.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**Section E: Security and Access Control**

**21. My documents and SK member records are secure in SKActHub.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**22. User authentication and role-based access control function properly.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**23. I have not observed any unauthorized access or potential security issues.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**Section F: Overall System Effectiveness**

**24. SKActHub has improved document management, activity monitoring, and SK member management.**

☐ 1 – Strongly Disagree

☐ 2 – Disagree

☐ 3 – Neutral

☐ 4 – Agree

☐ 5 – Strongly Agree

**25. Please provide any suggestions or comments to further improve SKActHub:**

**PRE-SURVEY QUESTIONNAIRE**

**Title:** SKActHub: A Project Management for Monitoring and Tracking SK Activities to Strengthen SK Governance for the Ormoc City Youth Development Office

**Purpose:** This survey aims to assess the existing document management practices, challenges, and system needs of the Sangguniang Kabataan (SK) offices prior to the implementation of SKActHub.

**Part I: Respondent Profile**

1. Name (Optional): ___________________________
2. Position:
   ☐ SK Chairperson
   ☐ SK Kagawad
   ☐ SK Secretary
   ☐ SK Treasurer
   ☐ SK Staff
3. Barangay: ___________________________
4. Years in Service:
   ☐ Less than 1 year
   ☐ 1–2 years
   ☐ 3–5 years
   ☐ More than 5 years

**Part II: Current Document Management Practices**

**Instruction:** Please select the answer that best describes your current situation.

1. How are SK documents currently stored?
   ☐ Purely paper-based
   ☐ Paper-based with some digital copies
   ☐ Mostly digital
   ☐ Fully digital
2. How often do you experience difficulty locating important documents?
   ☐ Never
   ☐ Rarely
   ☐ Sometimes

☐ Often

☐ Always

3. How long does it usually take to retrieve a specific document?

   ☐ Less than 5 minutes

   ☐ 5–15 minutes

   ☐ 16–30 minutes

   ☐ More than 30 minutes

4. Have you experienced lost or misplaced documents?

   ☐ Never

   ☐ Once

   ☐ A few times

   ☐ Frequently

5. How are membership records managed?

   ☐ Manual logbook

   ☐ Excel or spreadsheet

   ☐ Online system

   ☐ No organized system

**Part III: Monitoring and Reporting**

6. How do you track SK activities and events?

   ☐ Notebook/logbook

   ☐ Printed reports

   ☐ Spreadsheet

   ☐ No formal tracking system

7. How easy is it to generate reports for submission to local authorities?

   ☐ Very Easy

   ☐ Easy

   ☐ Neutral

   ☐ Difficult

   ☐ Very Difficult

8. Do you have a centralized system for monitoring project status?

   ☐ Yes

   ☐ No

9. How often do delays occur in document submission or reporting?

   ☐ Never

   ☐ Rarely

   ☐ Sometimes

☐ Often
☐ Always

**Part IV: Communication and Accessibility**

10. How do SK officials usually communicate regarding activities?
☐ Face-to-face
☐ Messenger/Chat apps
☐ SMS
☐ Email
☐ Written memos

11. Is there a centralized messaging system for official communication?
☐ Yes
☐ No

12. How accessible are important SK documents when needed?
☐ Very Accessible
☐ Accessible
☐ Moderately Accessible
☐ Hard to Access
☐ Not Accessible

**Part V: System Needs and Readiness**

13. Do you believe a digital document management system is necessary?
☐ Strongly Agree
☐ Agree
☐ Neutral
☐ Disagree
☐ Strongly Disagree

14. Which features would be most beneficial? (Select all that apply)
☐ Event management
☐ Member record management
☐ Messaging system
☐ Document storage
☐ Notifications
☐ Analytics dashboard

15. Are you willing to adopt a centralized digital system for SK operations?
☐ Yes
☐ No
☐ Maybe

# LIAN EMMANUEL V. BUGTAI

## CONTACT


+639513641101

lianbugtai@gmail.com

Bryg. Ipil, Osco Compound


## EDUCATION

**2022- Present**

**ACLC COLLEGE OF ORMOC**

- Bachelor of Science in Information Technology

**2020 - 2022**

**SAINT'S PETERS COLLEGE OF ORMOC**

- Humanities and Social Sciences (HUMSS)

**2016 - 2020**

**SAINT'S PETERS COLLEGE OF ORMOC**

**2010 - 2016**

**IPIL CENTRAL SCHOOL**

## PROFILE

A motivated IT student with a strong foundation in computer systems, front-end programming, and networking principles. Demonstrated ability to collaborate in team projects and manage tasks using leadership and time management skills. Familiar with HTML, CSS, and basic troubleshooting. Eager to gain hands-on experience through an IT internship and contribute to real-world projects.

## SKILLS

- Fast- Learner
- Teamwork
- Good Communication
- Critical Thinking
- Front-End Programming
- Microsoft Office Suite (Word, Excel, PowerPoint)
- Basic Python

## PROJECTS

**Freelance Marketplace System (School Project)**
Tools used: M.E.R.N. Stack
Collaborated with a team to develop a web-based platform that connects freelancers with job posters. The system allows users to create accounts, search for job listings, apply for jobs, and manage ongoing work through a user-friendly dashboard.

- Implemented user authentication for both freelancers and employers.
- Designed job search and filtering features to enhance usability.
- Built job posting, application, and status-tracking functionalities.
- Focused on responsive UI and intuitive navigation to ensure accessibility across devices.

## REFERENCE

**Engr. Ma. Lelena Trinidad**
OJT Adviser
ACLC College of Ormoc City, Inc.
Phone: +639703789767

**Ellen M. Guinares**
IT Instructor
ACLC College of Ormoc City, Inc
Phone: +639619713200

# LOUISE CZAR I. AWIT


Talisay, Matag-ob, Leyte

09935872560 / 09561293276

awitlouise16@gmail.com


## ABOUT ME

I am an enthusiastic and goal-driven Bachelor of Science in Information Technology student with a genuine interest in programming, system development, and solving technical challenges. With a foundation in software development, networking concepts, and database handling, I am excited to immerse myself in hands-on experiences. Recognized for being a quick learner, and flexible. I am dedicated to continuous improvement and eager to make meaningful contributions to a forward-thinking company during my on-the-job training.

## PROFILE

| | |
|---|---|
| **Age:** | 22 years old |
| **Birthdate:** | 05-12-2003 |
| **Sex:** | Male |
| **Civil Status:** | Single |

## PERSONAL SKILLS

- Strong communication and interpersonal skills
- Adaptable and quick learner
- Problem-solving mindset
- Strong Attention to Detail
- Goal-Oriented
- Time management and organizational skills

## TECHNICAL SKILLS

- Computer Literate
- Basic knowledge in programming languages
- Familiar with web development fundamentals
- Troubleshooting and basic computer hardware/software maintenance
- Understanding of database concepts

## EDUCATION

- **ACLC COLLEGE OF ORMOC**

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

2022-PRESENT

- **LA CONSOLACION COLLEGE-LILOAN**

SCIENCE TECHNOLOGY ENGINEERING MATHEMATICS: SENIOR HIGH SCHOOL

2020-2022

- **LA CONSOLACION COLLEGE-LILOAN**

JUNIOR HIGH SCHOOL

2016-2020

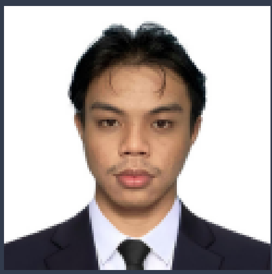
## LANGUAGE

- English
- Tagalog
- Bisaya

## REFERENCES

**ENGR. MA. LELENA TRINIDAD**
OJT Adviser
ACLC College of Ormoc City, Inc.
Ormoc City
09703789767

**ELLEN M. GUIÑARES**
IT Instructor
ACLC COLLEGE OF ORMOC CITY, INC.
09619713200



# JOHNLOE G. ARCILLAS

ACLC COLLEGE OF ORMOC

with [X]+ years of experience in developing and executing data-driven marketing strategies that increase brand visibility, customer engagement, and revenue. Skilled in digital marketing, content creation, campaign management, and market research. Proven track record of delivering ROI-focused solutions and collaborating across cross-functional teams to drive business growth.

## Contact

**Phone**
0962-602-6020

**Email**
rollslawie@gmail.com

**Address**
Brgy. Mabato, Ormoc City

## Education

2018 - 2020 (Junior High)
**Ormoc Immaculate Conception School Foundation Inc.**

2021 - 2022 Senior High
**San Lorenzo Ruiz College of Ormoc Inc.**

2022 - Present
**ACLC College of Ormoc Inc.**

## Skills

- Graphic Designing
- Video Editing
- Documenting

## Computer Skills

- Text Processor
- Slide Presentation

## Tools Used

- Microsoft Word
- Capcut
- Canva

## Language

**English**

## Experience

**2022 - 2023**
ACLC College of Ormoc
**Department Secretary**

- Accurately record, transcribe, and distribute meeting minutes for departmental and faculty meetings, maintaining a clear and organized record of discussions, decisions, and action items.
- Serve as the primary administrative support for the department head, ensuring smooth daily operations and efficient communication within the department.

**2024 - 2025**
Brgy. Mabato, Ormoc City
**SK Chairperson**

- Led the planning, implementation, and evaluation of various youth development programs, community outreach initiatives, and capacity-building activities in alignment with local government priorities.
- Oversaw the execution of youth-focused events including sports tournaments, clean-up drives, leadership training, and educational seminars.
- Maintained accurate documentation of projects, financial reports, and official resolutions for audit and compliance purposes.

**Other Related Skills**

- Photography
- Basic Graphic Designing (Canva)
- Video Creation

## Seminars Attended

- SK Planning, Budgeting and Review Process Training Workshop
- Training on Foresight and Anticipatory Leadership in Local Governance and Development

## Reference

**ENGR. MA. LELENA TRINIDAD OJT Adviser**
ACLC College of Ormoc City, Ormoc City
Phone: 0970 - 378 - 9767

**MRS. ELLEN GUIÑARES**
Adviser of Computer Studies Department
Phone: 0961 - 971 - 3200